



Weakly supervised segmentation of retinal layers on OCT images with AMD using uncertainty prototype and boundary regression

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ARTICLE INFO

Keywords:

Optical coherence tomography
Layer segmentation
Multi-scale fusion
Uncertainty prototype
Boundary regression

ABSTRACT

Retinal layer segmentation for optical coherence tomography (OCT) images of eyes is a critical step in the diagnosis and treatment of age-related macular degeneration (AMD) eye disease. In recent years, dense annotation supervised OCT layer segmentation methods have made significant progress. However, obtaining pixel-by-pixel labeled masks from OCT retinal images is time-consuming and labor-intensive. To reduce dependence on dense annotations, this paper proposes a novel weakly supervised layer segmentation method with Uncertainty Prototype module and Boundary Regression loss (W-UPBR), which only requires scribble annotations. Specifically, we first propose a feature enhancement U-Net (FEU-Net) to alleviate the severe layer distortion problem in OCT images with AMD. And this model serves as the backbone of a dual-branch network framework to enhance features. Within FEU-Net, in addition to the basic U-Net, two modules have been proposed: the global-local context-aware (GLCA) module, which captures both global and local contextual information, and the multi-scale fusion (MSF) module, designed for fusing multi-scale features. Secondly, we propose an uncertainty prototype module that combines the uncertainty-guided prototype and distance optimization loss. This module aims to exploit the similarities and dissimilarities between OCT images, thereby reducing mis-segmentation in layers caused by interference factors. Furthermore, a mixed pseudo-label strategy is incorporated to mix different predictions to alleviate the limitations posed by insufficient supervision and further promote network training. Finally, we design a boundary regression loss that constrains the boundaries in both 1D and 2D dimensions to enhance boundary under the supervision of generated mixed pseudo-labels, thereby reducing topological errors. The proposed method was evaluated on three datasets, and the results show that the proposed method outperformed other state-of-the-art weakly supervised methods and could achieve comparable performance to fully supervised methods.

1. Introduction

Age-related macular degeneration (AMD) is a retinal disease that affects people over the age of 55. With aging populations and longer life spans, AMD is an increasingly important public health problem. It is estimated that 8.69 % of the global population suffers from AMD, with the prevalence expected to increase to 288 million by 2040 (Guymer and Campbell, 2023). There are two main types of AMD: dry (atrophic) and wet (neovascular or exudative) (Taylor, 2012). Most AMD patients suffer from dry AMD, in which drusen is one of the early symptoms of AMD (Gheorghe et al., 2015; Kaarniranta et al., 2023). Typical drusen

appear as focal whitish yellow excrescences deep to the retina, while geographic atrophy (GA) represents the late stage of dry AMD, characterized by the loss of photoreceptors, retinal pigment epithelium (RPE), and underlying choriocapillaris (Fleckenstein et al., 2018). Although dry AMD is more common than wet AMD, wet AMD is the leading cause of vision loss and mainly includes pigment epithelial detachment (PED) and choroidal neovascularization (CNV) (Gheorghe et al., 2015). The former may be caused by the coalescence of serous fluid, fibrovascular tissue, hemorrhage, or drusen beneath the RPE, while the latter, known as wet late-stage AMD, manifests as abnormal blood vessels growing in the macular area beneath the retina. This may lead to leakage, bleeding

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<https://doi.org/10.1016/j.media.2025.103572>

Received 2 September 2024; Received in revised form 16 February 2025; Accepted 27 March 2025

Available online 29 March 2025

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and retinal damage, ultimately resulting in severe vision loss (Pour et al., 2023). Images of the four typical lesions belonging to the two types of AMD are shown in Fig. 1.

Optical Coherence Tomography (OCT) is a high-resolution, non-invasive imaging technology capable of capturing micron-scale structures within the human retina (Pekala et al., 2019). It is now widely used in ophthalmology, and previous studies (Fazekas et al., 2022; Xiang et al., 2018) have demonstrated that accurately quantifying layer thickness in OCT images can aid doctors in understanding the progression and severity of diseases. However, manually annotating a large number of OCT images is very time-consuming and the inefficient. Additionally, layer boundaries are more difficult to accurately differentiate in images with severe deformation, such as those in the middle and late stages of AMD. This complexity inevitably increases the difficulty of labeling. Hence, an automated retinal layer segmentation algorithm is required to assist clinicians in the diagnosis and monitoring of these eye diseases.

Initial layer segmentation algorithms can be roughly divided into two categories: algorithm-driven methods and machine learning-based methods. Algorithm-driven methods typically build mathematical models by utilizing prior information of images, such as intensity based methods (Ishikawa et al., 2005; Shahidi et al., 2005), A-scan based methods (Koozekanani et al., 2001) and graph based methods (Dufour et al., 2012; Kaba et al., 2015). However, these methods are more dependent on the quality of the data, are complex to implement, and have low robustness. Machine learning-based methods typically involve extracting hand-crafted features from images, which are then used to train a classifier (e.g., support vector machine (SVM) (Fuller et al., 2007), random forest (Lang et al., 2013; Xiang et al., 2018)) to assign each pixel to a specific class. However, generating manual features required by these methods can be challenging, and some post-processing techniques require adjustment of numerous parameters and exhibit poor generalization capabilities.

With the development of deep learning, deep learning methods based on convolutional neural network (CNN) have also been widely used in retinal layer segmentation tasks and have achieved great progress and success. In (Fang et al., 2017), a CNN classifier was trained by extracting patches to classify layer boundaries and further optimized the boundaries through post-processing methods to segment nine layer boundaries in OCT images of AMD patients. Differently, (Roy et al., 2017) proposed an end-to-end fully convolutional network to segment seven retinal layers and fluid pixel by pixel. The former patch-based classification method has high computational cost and low efficiency, while the latter, employing pixel-by-pixel classification, suffers from layer topology errors and boundary deviations. To further improve the segmentation accuracy, (Liu et al., 2020) proposed using column attention to focus on the affinity between each pair of pixels on A-scan (each A-scan represents a column of a B-scan image, containing a complete retinal layer structure). This integration of long-range dependencies aims to achieve global information perception effects. However, it overlooks the interaction between columns. Refs. (He et al., 2021; Wang et al., 2021a) and (Kepp et al., 2019; Liu et al., 2020) further optimized layer segmentation results by ensuring the topological order

of retinal layers. They achieved this by modeling layer boundary coordinates and using distance map for boundary regression, respectively. However, they need to use dense annotations supervision to achieve better performance.

Apart from the methods mentioned above, most of the successful layer segmentation methods currently rely on a large number of dense annotations. However, the workload of clinicians, complex image structures, and high requirements for professional knowledge make obtaining such dense annotations a significant challenge. Moreover, creating dense annotations involves manual annotation pixel by pixel, which is both expensive and time-consuming. On the other hand, although previous work has considered the use of a priori information such as topology, it overlooked the similarities and dissimilarities between OCT images. Specifically, there are similarities in inherent features such as layer structure between OCT images, but dissimilarities in interference factors such as noise and artifacts across different images. An example of a B-scan with artifacts is illustrated in Fig. 1(a) and circled in red.

Although previous layer segmentation methods have achieved some success, we observed some shortcomings: (1) Most existing methods require dense annotations, which are difficult, time-consuming and labor-intensive to obtain. (2) OCT retinal images of patients with AMD generally exhibit severe layer distortion, particularly in the middle and late stages of the disease. As shown in Fig. 1(c) and (d), the RPE will detach from the Bruch's Membrane (BM) and bulge or abnormal blood vessels will destroy the layer structure. However, most existing methods do not consider this problem, making it difficult to perceive the lesion region, resulting in poor segmentation results. (3) The structures within OCT images are similar, but there are also dissimilarities such as noise and artifacts between different B-scans. However, most current methods do not explicitly take this prior information into account.

To alleviate these issues, we propose a novel weakly supervised retinal layer segmentation framework using only scribble annotations, called **W-UPBR**. This framework is based on a dual-branch network architecture, with a backbone network designed as a feature enhancement U-Net (FEU-Net). The Uncertainty Prototype module, and Boundary Regression loss are also incorporated into the framework. To reduce the workload of pixel-level labeling, we choose to use scribble annotations instead of dense annotations for OCT retinal layer segmentation. Fig. 2(b) and (c) illustrate examples of dense annotation and scribble annotation of a B-scan in the Cell-AMD dataset (Kermayn et al., 2018), respectively. Compared with dense annotations, scribble annotations can significantly reduce annotators' workload. In addition, scribble annotations can provide richer supervision information when compared to image-level labels (Liu et al., 2023) and point labels (Laradji et al., 2021). Moreover, labeling scribble annotations is relatively flexible and more suitable for complex retinal layer structures. Therefore, we chose scribble annotations as the supervision signal for this work. For severe retinal layer deformation problems in AMD patients, we propose a global-local context-aware module. This module addresses the distortion caused by lesions in the retinal layer structure by capturing global and local contextual information. This module calculates the affinity between every two pixels in each column of the OCT

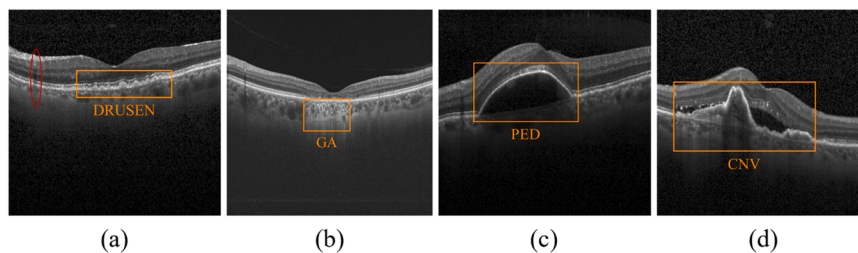


Fig. 1. Examples of four lesion images of two types of AMD, with biomarkers outlined in orange. (a) drusen; (b) geographic atrophy (GA); (c) pigment epithelial detachment (PED); (d) choroidal neovascularization (CNV). In addition, artifacts are also circled in red in (a).

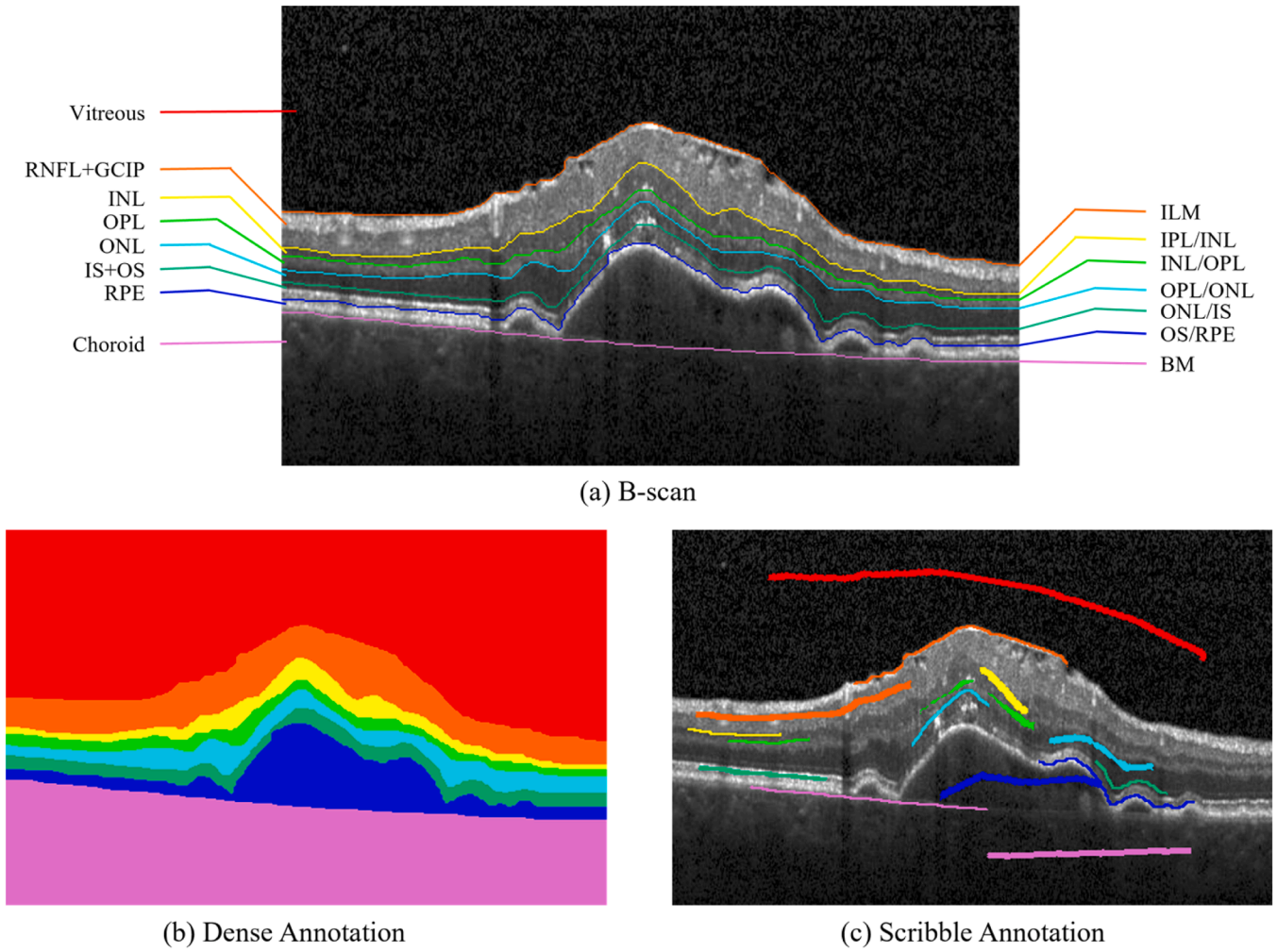


Fig. 2. (a) An example OCT B-scan image in Cell-AMD dataset. There are eight regions marked on the left side of the image, including two background regions (Vitreous and Choroid) and six layer regions, and seven layer boundaries are marked on the right side. The seven layer boundaries from top to bottom are: ILM (Internal Limiting Membrane), IPL/INL (Inner Plexiform Layer/Inner Nuclear Layer), INL/OPL (Inner Nuclear Layer/Outer Plexiform Layer), OPL/ONL (Outer Plexiform Layer/Outer Nuclear Layer), ONL/IS (Outer Nuclear Layer/Inner Segment), OS/RPE (Outer Segment/Retinal Pigment Epithelium), and BM (Bruch's Membrane). (b) Dense annotation for the B-scan. (c) Scribble annotation for the B-scan.

image through the column attention module (Liu et al., 2020) to obtain global context information. On this basis, we introduce dilation convolutions with different dilation rates to perceive layer regions under different receptive fields to obtain local information. At the same time, this approach addresses the shortcoming of the column attention module, which ignores the interaction between columns. Additionally, we propose a multi-scale fusion module to merge the texture and spatial information of shallow features with the semantic information of deep features, resulting in output features with enhanced information content. To reduce the impact of interference factors on prediction accuracy, we introduce the concept of prototypes. By capturing the similarities and dissimilarities between B-scans, we aim to reduce the mis-segmentation of pixels in layers caused by factors such as noise, artifacts, addressing the intra-layer false positive problem. Our approach projects the features from multiple B-scans into the embedding space, generates prototypes representing specific classes through prototype learning, and optimizes the distance between feature vectors. This involves bringing feature vectors of the same categories closer together and pushing those of different categories farther apart using distance optimization loss. To generate more representative prototypes under the supervision of scribble annotations, we filter embedded features by calculating uncertainty, excluding those with high uncertainty.

We also observed some challenges in OCT retinal layer segmentation

supervised by scribble annotations. First, using only scribble annotation supervision limits network performance. In previous work on medical images, several strategies have been proposed to alleviate this problem, including consistency regularization (Liu et al., 2022), mix augmentation techniques (Zhang and Zhuang, 2022), and dynamically mixed pseudo labels (Luo et al., 2022). Secondly, for a layer segmentation task, it is crucial to obtain accurate, continuous, and smooth layer boundaries (Ngo et al., 2019). Scribble annotations lack complete annotations and cannot be directly used to supervise boundary learning. To address the challenges observed above, we first adopt a strategy of mixed pseudo-label generation to extend scribble annotations. It provides additional supervision information to assist network training. This strategy can also compensate for the weaknesses of individual predictions while improving the diversity of pseudo-labels (Luo et al., 2022). Furthermore, under the supervision of the pseudo-labels, we design a boundary regression loss. This loss function constrains boundaries in both 1D and 2D dimensions to enhance boundary learning and mitigate topological errors.

In summary, the main contributions are as follows:

- (1) We propose a weakly supervised retinal layer segmentation method with scribble annotations, named W-UPBR. To the best of our knowledge, it is the first exploration of using scribble

supervision for the retinal layer segmentation task. It employs a dual-branch architecture with FEU-Net serving as its backbone network. In the FEU-Net, a global-local context-aware (GLCA) module is proposed to extract both global and local context information, while a multi-scale fusion (MSF) module to fuse shallow and deep features to enhance the output features.

- (2) To alleviate the mis-segmentation in layers caused by interference factors, we propose an uncertainty prototype module to leverage the similarities and dissimilarities between OCT images. This module employs uncertainty estimation to guide prototype updates, thereby ensuring the representativeness of prototypes supervised by scribble annotations. Furthermore, distance optimization loss is introduced to optimize the intra- and inter-class feature distances.
- (3) The sparsity of scribble annotations limits their direct application in boundary regression tasks. Therefore, we design a joint boundary regression loss that combines 1D boundary loss and 2D reconstruction loss, and use pseudo labels generated by a mixed pseudo-label strategy for supervision, aiming to enhance boundary learning and thus mitigate topological errors.
- (4) We evaluated the proposed method on two datasets with AMD and one healthy dataset, comparing it with other state-of-the-art methods. The results demonstrated the superiority of the proposed method.

The remainder of this article is organized as follows. Related works are presented in Section 2. The detailed description of proposed method is given in Section 3. Section 4 shows the experimental results and we conclude the whole work in Section 5.

2. Related works

In this section, we review some related works, including: (1) layer segmentation methods, (2) weakly supervised segmentation in medical images, and (3) prototype learning.

2.1. Layer segmentation methods

- (1) **Traditional Methods:** Before deep learning became popular, the task of retinal layer segmentation was mainly achieved through some traditional algorithms and machine learning methods. Yang et al. (2010) proposed an edge detection algorithm based on gradient and shortest path to segment nine boundaries. They first collected gradient information by extracting local Canny edge maps and global axial intensity gradients, and then used the shortest path search algorithm to optimize the boundaries. Although the customized Canny edge detection can detect boundaries well, it is difficult to deal with noise interference. Chiu et al. (2010) proposed a fully automatic segmentation method that successfully segmented eight retinal layer boundaries in normal adults by using graph theory and dynamic programming. The above two methods use graph-based methods to segment and optimize layer boundaries, while graph-based methods are easily affected by image qualities and parameter settings. As a result, some researchers have turned their attention to machine learning methods to segment layer boundaries by learning and extracting features for pixel classification. Vermeer et al. (2011) used multi-scale gray values and gradients as feature vectors, defined based on A-scans. They employed Support Vector Machines (SVM) as a classifier to classify pixels and implemented the level set method to alleviate mis-segmentation. Chiu et al. (2015) introduced a classification method based on kernel regression. They utilized the GTDP algorithm framework proposed in (Chiu et al., 2010) to further optimize the retinal layer boundaries, thereby enabling the segmentation of seven retinal layers. Indeed, machine learning methods often rely on manually

designed features and may require additional pre-processing or post-processing techniques to optimize segmentation results. Consequently, when dealing with large-scale datasets and images with lesions, machine learning methods may encounter challenges.

- (2) **Deep Learning Methods:** Many recent methods are proposed based on deep convolutional neural networks. Fang et al. (2017) decomposed an OCT image of a patient with AMD into patches and input them into a CNN model to extract features. Subsequently, they used a modified GTDP algorithm to convert the CNN probability map into the final layer boundaries. Liu et al. (2018) trained a structured random forest classifier by integrating deep features with hand-designed features and used the shortest path algorithm to obtain the final layer segmentation. However, both patch-based methods entail a high computational burden and necessitate post-processing. Roy et al. (2017) proposed an end-to-end method called ReLayNet, which adopts a U-Net-like encoder-decoder architecture to perform pixel-wise classification to retinal layer segmentation. However, these methods are difficult to ensure the consistency of the topological order of retinal layers. To alleviate this problem, recent researchers have proposed some regression-based methods, which adopt two strategies: modeling boundary coordinates and distance map. He et al. (2021) and Wang et al. (2021a) modeled the distribution of surface positions and used a topology-preserving module to ensure the layer topological order. Unlike both methods of enforcing correct topology, Tan et al. (2023) proposed a lightweight architecture that combines CNN and Transformer. This architecture incorporates a boundary regression loss to correct boundary deviations by regressing coordinates and gradients, while also minimizing topological errors. Both Kepp et al. (2019) and Liu et al. (2020) utilized signed distance maps (SDM) for regression tasks, albeit with slightly different processing methods. The former is based on layer region masks, while the latter is based on layer boundary masks. Although these methods have demonstrated good performance, they typically rely on time-consuming and laborious manual pixel-by-pixel labeling. In contrast, we employ scribble annotations to supervise the OCT retinal layer segmentation task, significantly reducing the labeling workload.

2.2. Scribble annotation weakly supervised segmentation

Compared with dense annotation, weakly supervised annotation does not require annotating an image pixel by pixel, so it can reduce the burden of manual annotation. There are many forms of supervision signals for weakly supervised segmentation methods: image-level (Kuang et al., 2023; Liu et al., 2023), point (Laradji et al., 2021), scribble (Li et al., 2023), and bounding box (Mahani et al., 2022). Scribble annotation, as a weak supervision signal, has received widespread attention and application in the field of medical image segmentation because it can reduce the annotation burden while providing necessary supervision. Liu et al. (Liu et al., 2022) segmented scribble-annotated CT images using the mean teacher model, filtering out uncertain pixels through uncertainty estimation and correcting biases through consistency regularization. Instead of using scribble annotation supervision directly, Zhang and Zhuang (2022) designed a mix image enhancement method to add scribble annotations and optimize the model through cycle consistency losses, thereby achieving better performance on heart segmentation tasks. In addition, Luo et al. (2022) generated pseudo labels by dynamically mixing the outputs from two different decoders and further supervised the network training, thereby improving network performance.

The current weakly supervised methods based on scribble annotations have made great progress in medical image segmentation. These methods introduce technologies such as dual-branch or dual-path

networks, consistency regularization, and pseudo label, directly using scribble annotation for supervision or generating pseudo labels through scribble annotation for indirect supervision to enhance network performance. At the same time, it not only proves their feasibility but also demonstrates the possibility of reducing manual annotation. Although weakly supervised methods based on scribble annotation have been successfully applied to medical images and achieved good performance, there has been no relevant exploration and application in the OCT retinal layer segmentation task.

2.3. Prototype learning

Prototype-based learning is a machine learning method. The prototype is usually defined as the center of a representative sample or data point in a sample space (such as the mean vector). The nearest neighbor algorithm (Cover and Hart, 1967) and learning vectorization (LVQ) (Kohonen, 1990) are classic prototype-based methods in machine learning. The concept of prototype was introduced by researchers into many deep learning tasks. The core idea is to generate a representative prototype for each class and then calculate the distance to the embedded features to perform classification or segmentation tasks. Recently, networks based on prototype-based few-shot learning have become increasingly popular. For example, Snell et al. (2017) proposed a few-shot prototype network to perform classification tasks and extended it to zero-shot learning. Kwon et al. (2021) proposed a dual-prototype contrastive learning method for few-shot segmentation tasks to effectively capture representative semantic features. Different from (Snell et al., 2017), Kwon et al. introduced contrastive learning to increase the inter-class distances and reduce intra-class distances, thereby making prototypes more discriminative. In addition, the prototype idea has also been applied to the semantic segmentation task. Zhou et al. (2022) constructed a non-parametric segmentation framework that represents each class as a set of non-learnable prototypes. The update of the prototypes only relies solely on the average features of the training pixels in that class, and the final dense prediction is achieved by retrieving a set of non-learnable prototypes. Additionally, Xu et al. (2022) introduced the prototype concept into semi-supervised learning. Their method is based on the Cut-Mix operation and the student-teacher framework, which encourages consistency between the output from the prototype-based predictor in the student model and the prediction from the linear predictor to make intra-class feature representation more compact. Motivated by the above studies, we further extend the prototype-based learning task to the OCT retinal layer segmentation problem under scribble supervision. Specifically, we capture the similarities and dissimilarities between OCT images through multi-class prototypes and introduce uncertainty estimation to guide the prototype updates, generating more representative prototypes for dense predictions.

3. Method

In this section, we describe in detail the proposed method for weakly supervised retinal layer segmentation via scribble annotations on OCT images. We first outline the problem and present the proposed scribble-supervised segmentation framework. Subsequently, we elaborate on the core components and losses of the proposed method, including the backbone network referred as Feature Enhancement U-Net (FEU-Net) in the dual-branch network architecture, the uncertainty-guided prototype module and the boundary regression loss. Finally, we describe the loss functions used for network training.

3.1. Problem setup and overview of framework

In this work, our task is to assign a label of $\{0, 1, \dots, C-1\}$ to each pixel of an OCT retinal layer image, where C represents the number of categories. The inputs are an OCT image and the corresponding scribble

annotation. To better illustrate our method, we use $X \in \mathbb{R}^{H \times W}$ to denote the input image and $Y \in \{0, 1, \dots, C-1, 255\}^{H \times W}$ to denote the scribble annotation corresponding to image X , where H and W represent the height and width of the image respectively. Y includes $C+1$ categories of labels: 0 and $C-1$ for the top and bottom backgrounds pixels respectively, 1 to $C-2$ for scribbled pixels in the retinal layer regions, and 255 for the unannotated pixels. The number of layer boundaries B is $C-1$. In Cell-AMD dataset, C is set to 8. An example of OCT B-scan in the Cell-AMD dataset is shown in Fig. 2(a). Fig. 3 illustrates the detailed pipeline of the proposed W-UPBR. The proposed W-UPBR consists of a dual-branch network with the designed backbone network FEU-Net, an uncertainty prototype (UP) module, and a boundary regression loss (BRL). To be more specific, we first use a dual-branch framework based on scribble supervision for initial network training. This framework uses the designed FEU-Net as the backbone network to enhance features, thereby alleviating the layer distortion caused by lesions. Meanwhile, the uncertainty prototype module is designed to utilize the similarities and dissimilarities between B-scans to reduce mis-segmentations in layers. Then, the network is further optimized using generated mixed pseudo-labels instead of scribble annotations. In addition, under the supervision of the mixed pseudo-labels, we design a boundary regression loss from both 1D and 2D dimensions to enhance boundary learning and alleviate topological errors.

For a clearer explanation, we formalize the proposed weakly supervised layer segmentation method that utilize only scribble annotations as follows. In the training phase, we input the original image X into the main branch $B_m(\cdot)$ and the transformed image $T(X)$ into the transform branch $B_t(\cdot)$, respectively. The same transformation is also performed on the scribble annotation Y corresponding to the input in the transform branch to obtain $T(Y)$, where $T(\cdot)$ represents the transformation function. First, to optimize the outputs of the segmentation heads (i.e., P^s and P^t) from the two branches, we respectively compare them with the corresponding scribble annotations via partial cross-entropy loss ($\mathcal{L}_{pce}$). Then, the certainty pseudo-label Y^u and the generated hard prediction Y^s are fed into the uncertainty prototype module, both of which are output by the main branch. The former is used to guide the generation of certainty features, while the latter is used to supervise prototype updates through distance optimization losses, including one relative distance loss ($\mathcal{L}_{rdl-ce}$) and two absolute distance losses ($\mathcal{L}_{adl-cl}$, $\mathcal{L}_{adl-com}$). The output P^{pro} of the uncertainty prototype module is a similarity map, which is obtained by calculating the cosine similarity between the prototype and the embedded features. Subsequently, we use the mean square error loss ($\mathcal{L}_{mse}$) to minimize the difference between $T^{-1}(P^t)$ and P^s , as well as the difference between $T^{-1}(P^t)$ and P^{pro} . Here, $T^{-1}(P^t)$ is obtained by applying the inverse transform on the output of the transform branch. Furthermore, to alleviate the limitation of lack of supervision, the mixed pseudo-label Y^{mix} is generated by combining the output from the uncertainty prototype module and the inverse transformed output of the transform branch. This mixed pseudo-label is then used to perform the dice loss ($\mathcal{L}_{dice}$) to further optimize the network. Finally, under the supervision of the mixed pseudo-label, the cross-entropy loss ($\mathcal{L}_{CE}$) is used to initially optimize the output of the regression head from the main branch P^r . Then, the boundaries are constrained from 1D and 2D dimensions through two losses, $\mathcal{L}_{reg-1D}$ and $\mathcal{L}_{reg-2D}$, respectively. In the testing phase, we only use FEU-Net to predict the retinal layer segmentation results in an OCT image.

3.2. FEU-Net

Although scribble annotations can reduce the workload of labeling, its small number of annotated pixels affects the network's ability to learn global information, thereby limiting the network performance. Recent studies (Laradji et al., 2021; Liu et al., 2022) have shown that a dual-branch network architecture based on transformation consistency is effective in weakly supervised medical image segmentation. This

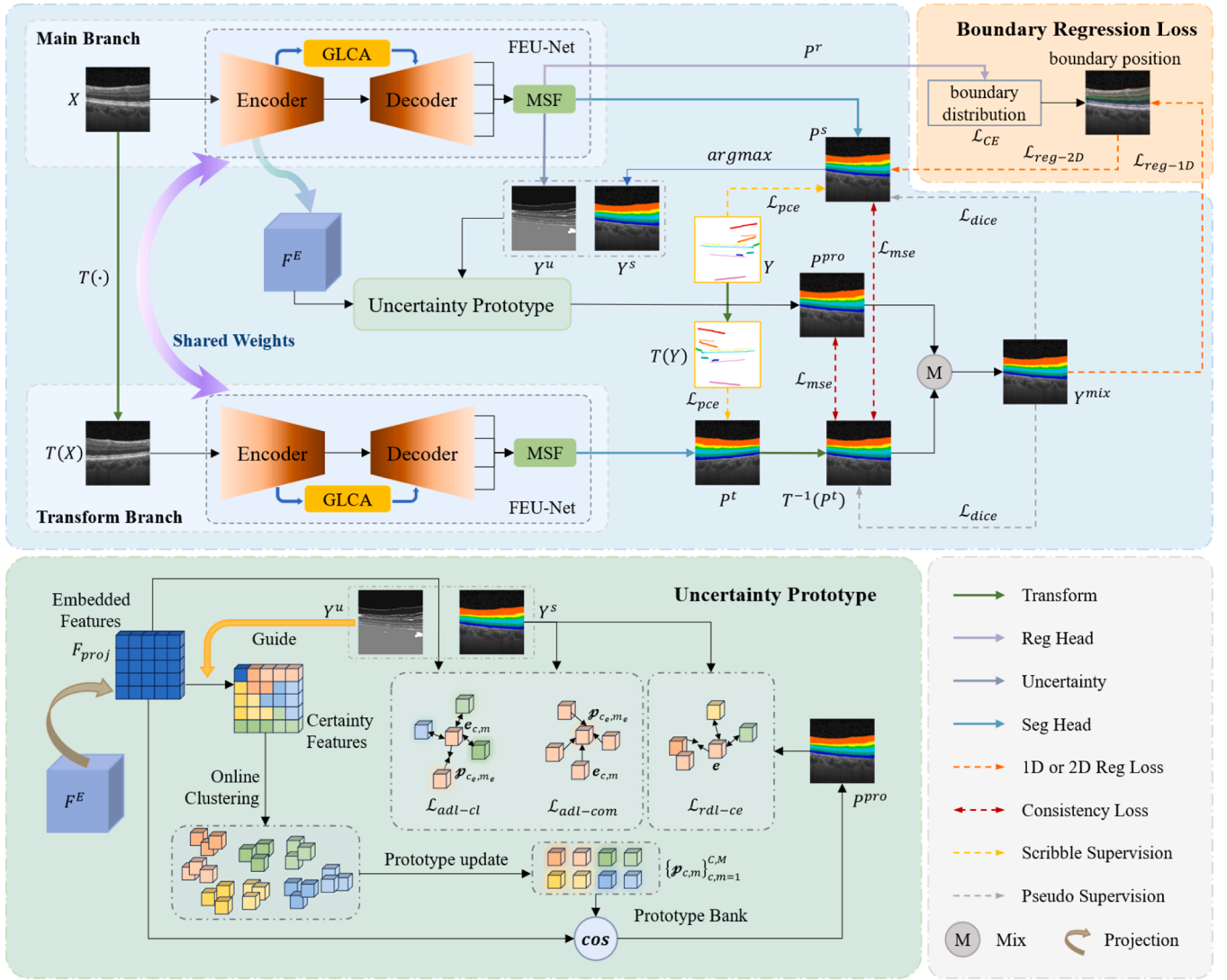


Fig. 3. Overview architecture of our proposed W-UPBR. The framework adopts a dual-branch network architecture with shared weights, using FEU-Net as its backbone. In addition, the framework includes an Uncertainty Prototype (UP) module and Boundary Regression Loss (BRL). FEU-Net consists of U-Net and two designed modules, including Global-Local Context-Aware (GLCA) module and Multi-Scale Fusion (MSF) module. The proposed method contains five loss terms: the scribble supervision loss ($\mathcal{L}_{pce}$) and pseudo-supervision loss ($\mathcal{L}_{dice}$) are used to supervise the segmentation predictions of the two branches, while the consistency loss ($\mathcal{L}_{mse}$) is used to minimize the difference between the segmentation prediction of the transform branch and the segmentation prediction of the main branch, and also minimize the difference between the segmentation prediction of the transform branch and the prototype prediction. Furthermore, the distance optimization loss in the uncertainty prototype module, including relative distance loss ($\mathcal{L}_{rdl-ce}$) and absolute distance loss ($\mathcal{L}_{adl-cl}$ and $\mathcal{L}_{adl-com}$), is used to optimize the inter- and intra-class distances. Finally, the boundary regression loss consists of $\mathcal{L}_{CE}$, $\mathcal{L}_{reg-1D}$ and $\mathcal{L}_{reg-2D}$, which enhance boundary learning by supervising the regression and segmentation prediction of the main branch.

architecture combines the idea of self-supervised learning and encourages the consistent prediction results under perturbation conditions. It not only improves the model's robustness, but also strengthens the performance of the main branch through the transform branch, thereby mitigating the effects of limited supervision. Based on these advantages, we also proposed a dual-branch network architecture (referred to as the main branch and the transform branch respectively) as shown in Fig. 3. The output of this architecture consists of different prediction probability maps, which are obtained by feeding the original and transformed images into the shared weight backbone network FEU-Net. To initially train the network, we applied the scribble supervision loss (Eq. (21), details provided in Section 3.5) and consistency loss (Eq. (22), details provided in Section 3.5) separately on the two branches. Next, we will introduce the proposed backbone network FEU-Net, in detail.

In OCT images with AMD, the retinal layer may be severely distorted due to the presence of pathological biomarkers, especially in the OCT

images of patients with intermediate and late stages (Fang et al., 2017; Fazekas et al., 2022). As illustrated in Fig. 4, to mitigate the impact of lesion-induced retinal layer distortion on the OCT retinal layer segmentation task, we designed the FEU-Net. The proposed network consists of a U-Net and two proposed modules. Specifically, based on the U-Net architecture, we add the proposed GLCA module on the skip connection to deliver the extracted global and local context information to the decoder. In the decoder, we use a proposed MSF module to integrate the multi-scale information of the decoder to obtain the final output.

(1) Global-Local Context-aware: Faced with layer distortions like those in Fig. 1(c) and (d), a relatively large receptive field helps to localize the lesion, which can mitigate the impact caused by local distortions (Tan et al., 2023). Furthermore, the OCT B-scan is composed of a series of A-scans, each A-scan has a complete layer

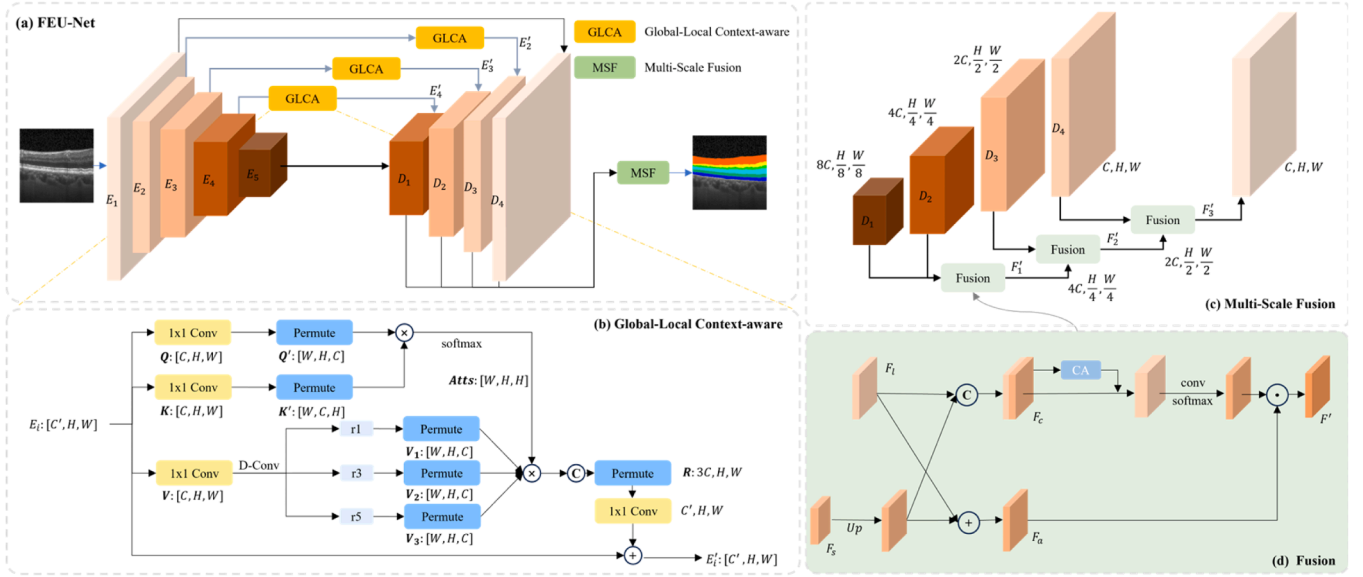


Fig. 4. Illustration of the designed feature enhancement U-Net (FEU-Net). (a) The overall architecture of FEU-Net. (b) Global-Local Context-aware (GLCA) module. (c) Multi-scale Fusion (MSF) module, which contains a Fusion module (d).

structure. Therefore, long-range dependencies can be integrated by calculating the affinity between every two pixels in the same A-scan (Liu et al., 2020). Based on this idea, we proposed a global-local context-aware (GLCA) module. It captures global information through column attention. Additionally, dilation convolutions are used to obtain different receptive fields and filter out some redundant information, which has been proven to be effective in dealing with such layer distortions (Apostolopoulos et al., 2017). In other words, the introduction of dilation convolution can enhance the interaction between nearby A-scans to perceive the lesion region.

As shown in Fig. 4(b), the module first permutes the dimensions of the two features Q and K to obtain $Q' \in \mathbb{R}^{W \times H \times C}$ and $K' \in \mathbb{R}^{W \times C \times H}$. Then, a matrix multiplication in the W dimension is performed between Q' and K' , followed by a *softmax* to calculate the column attention map $Atts \in \mathbb{R}^{W \times H \times H}$. Meanwhile, we use dilation convolution with dilation rates of $[1, 3, 5]$ on feature V to obtain three new features $\{V_1, V_2, V_3\} \in \mathbb{R}^{W \times H \times C}$ with different receptive fields. Consequently, we perform matrix multiplication between $Atts$ and V_1, V_2 and V_3 respectively, then concatenate the products and permute the result to $R \in \mathbb{R}^{3C \times H \times W}$. Finally, for the input feature $E_i \in \mathbb{R}^{C \times H \times W}$ ($i = \{2, 3, 4\}$) in the i th stage from the encoder, its output feature $E'_i \in \mathbb{R}^{C \times H \times W}$ through this module can be computed by

$$E'_i = pReLU(Conv(Perm(Concat(Atts \otimes V_1, Atts \otimes V_2, Atts \otimes V_3))) + E_i) \quad (1)$$

where $Conv(\cdot)$ and $Perm(\cdot)$ represent 1×1 convolution and permutation operation respectively. $Concat(\cdot)$ is the concatenation operation, $\otimes$ represents the matrix multiplication.

(2) Multi-scale Fusion: Low-level features contain more texture information, such as boundary information, while high-level features have more abstract and rich semantic information. By fusing these two features, we can obtain features with richer information, which is helpful for OCT retinal layer segmentation. However, due to the semantic and scale differences between features at different levels, directly sampling and concatenating features will limit the representational capacity of multiscale features (Sun et al., 2023). Therefore, we proposed a multi-scale fusion (MSF) module, as shown

in Fig. 4(c). This module consists of three fusion stages $j(j = 1, 2, 3)$ to fuse features at different scales. The four features output by the

decoder are expressed as $\left\{ D_i \in \mathbb{R}^{2^{4-i}C \times \frac{H}{2^{4-i}} \times \frac{W}{2^{4-i}}}, i = 1, 2, 3, 4 \right\}$, where

H and W are the height and width of the input image respectively. C represents the dimension of the feature, and $C = 16$. The workflow of MSF is described by the following mathematical formula:

$$F'_1 = Fusion(D_1, D_2) \quad (2a)$$

$$F'_2 = Fusion(F'_1, D_3) \quad (2b)$$

$$F'_3 = Fusion(F'_2, D_4) \quad (2c)$$

where F'_j is the feature output by the j th fusion stage and $j = \{1, 2, 3\}$.

Furthermore, the fusion module is shown in Fig. 4(d). Specifically, for the j th stage of fusion, the fusion module first up-samples the small-size feature $F_s \in \mathbb{R}^{2^{4-j}C \times \frac{H}{2^{4-j}} \times \frac{W}{2^{4-j}}}$ to the same size as the large-size feature $F_l \in \mathbb{R}^{2^{3-j}C \times \frac{H}{2^{3-j}} \times \frac{W}{2^{3-j}}}$, then performs concatenation and addition operations on these two features respectively to obtain $F_c \in \mathbb{R}^{2^{4-j}C \times \frac{H}{2^{3-j}} \times \frac{W}{2^{3-j}}}$ and $F_a \in \mathbb{R}^{2^{3-j}C \times \frac{H}{2^{3-j}} \times \frac{W}{2^{3-j}}}$. Then F_c is input into the channel attention (CA) module (Hu et al., 2018) to output the channel attention map, and is multiplied with the original input feature F_c to obtain the weighted feature. Finally, it is fed into the convolution and *softmax* operation, and its output is element-wise multiplied with F_a to obtain the final output feature $F'_j \in \mathbb{R}^{2^{3-j}C \times \frac{H}{2^{3-j}} \times \frac{W}{2^{3-j}}}$ of the fusion module. Eqs. (3a)–(3c) describe the fusion module.

$$F_a = F_l + Up(F_s) \quad (3a)$$

$$F_c = Concat(F_l, Up(F_s)) \quad (3b)$$

$$F' = softmax(Conv(CA(F_c))) \cdot F_a \quad (3c)$$

where $Up(\cdot)$ represents the $2 \times$ bilinear up-sampling operation, and $Conv(\cdot)$ is the convolution operation.

3.3. Uncertainty based prototypes

In prototype learning, we typically learn prototypes for corresponding categories and then match a test sample with the prototype at the pixel level to perform pixel-by-pixel prediction (Dong et al., 2022). Recent studies (Zhou et al., 2022) show that a class can be represented using a set of prototypes, thus well capturing class characteristics and intra-class variance. In addition, contrastive learning is often introduced to optimize inter- and intra-class distances, thereby enhancing feature discriminability across images (He et al., 2022; Wang et al., 2021b). For OCT images, the retinal layer structure between B-scans shows high similarity, while there are differences in noise, artifacts, etc., between different B-scans. However, previous methods (Fazekas et al., 2022; Lu et al., 2023) did not explicitly consider exploiting this information. Therefore, to alleviate the mis-segmentation phenomenon caused by interference factors during layer segmentation, an uncertainty prototype module has been designed based on prototype learning and uncertainty estimation. Specifically, we introduce the concept of multi-class prototypes to capture class features. Through the distance optimization loss (Zhou et al., 2022), the similarities and dissimilarities between OCT images are further optimized. Additionally, considering that our proposed method is based on scribble supervision, we generate a certainty pseudo-label through uncertainty estimation to further filter out features with high uncertainty. This process guides the update of the prototype, thereby maintaining the representativeness and accuracy of the prototype. Finally, the output of the module is obtained by pixel-by-pixel matching of prototypes with embedded features. The uncertainty prototype module is depicted by the green block in Fig. 3.

(1) Certainty Pseudo Label Generation: Since scribble annotations present a large number of unlabeled pixels, predictions from a scribble-supervised model may generate large amounts of uncertainty for pixels. This may lead to the introduction of numerous embedded features with high uncertainty during the process of generating prototypes, thereby reducing the representativeness of the prototype. Therefore, to ensure the representativeness of the prototype, we first employ Monte Carlo Dropout (Yu et al., 2019) to estimate the uncertainty of the FEU-Net segmentation prediction. Uncertainty map $u \in \mathbb{R}^{H \times W}$ can be defined as (Yu et al., 2019):

$$u = -\sum_{c=1}^C p^c \log p^c \quad (4)$$

where $p \in \mathbb{R}^{C \times H \times W}$ is the mean of the probability maps of N forward propagations, and p^c is the mean probability map p of the c^{th} class. To balance estimation quality and training efficiency, N is set to 8.

Then, we generate certainty pseudo label Y^u through a threshold T , which is used to guide prototype generation. When the uncertainty is lower than the threshold T , it indicates that the certainty is high, so the certainty pseudo label Y^u is as follows:

$$Y^u = \begin{cases} \text{argmax}(p), & u < T \\ 255, & u \geq T \end{cases} \quad (5)$$

where $Y^u \in \{0, 1, \dots, C-1, 255\}^{H \times W}$, where $C = 8$.

(2) Prototype Generation and Update: A prototype can be a sample feature or a mean vector of sample features in the same category. The concept of multiple prototypes in a class introduced here obviously belongs to the latter. For the prototype bank in Fig. 3, we can define it as $\{\rho_{c,m} \in \mathbb{R}^D\}_{c,m=1}^{C,M}$, which represents the set of m^{th} prototypes of class c , and randomly initialize them. First, this module receives the features $\{E_2, E_3, E_4\}$ from the encoder $B_m^E(\cdot)$ of FEU-Net (see Fig. 4

(a)). It then bilinearly interpolates these features to the input image size and concatenates them, resulting in the feature $F^E \in \mathbb{R}^{C \times H \times W}$, where $C = 224$. Subsequently, we project it to obtain the embedded features $F_{\text{proj}} \in \mathbb{R}^{D \times H \times W}$, where $D = 128$.

In the case of using only scribble annotations, there is a problem that the update of the prototype relies on the corresponding embedding of the labeled pixels to ensure its representativeness. However, due to the existence of a large number of unlabeled pixels, the corresponding embeddings of these pixels are uncertain. Moreover, solely relying on scribble annotations to supervise the updating of prototypes results in a smaller number of available embeddings. As a result, the representativeness of the prototypes is adversely affected. Therefore, to ensure the representativeness of the prototypes, on the one hand, we use the certainty pseudo-labels Y^u obtained by Eq. (5) to guide the generation of certainty features $F'_{\text{proj}} \in \mathbb{R}^{D \times H \times W}$ by filtering out the embedded features with high uncertainty. On the other hand, we employ online clustering (Zhou et al., 2022) to assign embeddings with high certainty in a batch to update prototypes. Instead of scribble annotations, hard predictions Y^s are used to supervise the updating of prototypes, where Y^s can be obtained by performing an *argmax* operation on the output of P^s from the segmentation head of the main branch. This approach can further filter embeddings pixels while increasing the number of available embeddings. Subsequently, we use momentum to update the prototype, and the formula can be written:

$$\rho_{c,m}^{\text{new}} = \alpha \rho_{c,m}^{\text{old}} + (1 - \alpha) \bar{e}_{c,m} \quad (6)$$

where $\rho_{c,m}^{\text{old}} \in \mathbb{R}^D$ and $\rho_{c,m}^{\text{new}} \in \mathbb{R}^D$ represent the m^{th} prototype of the c^{th} class before and after the update, respectively, which can also be called the m^{th} subclass of the c^{th} class. D is the dimension of the prototype vector. α is a hyperparameter, which is set to 0.999 following (Zhou et al., 2022) and $\bar{e}_{c,m} \in \mathbb{R}^D$ represents the mean value of embedded training pixels of the m^{th} prototype belonging to the c^{th} class. Note that $\bar{e}_{c,m}$ is obtained by averaging the highly certainty embedded pixels $e_{c,m}$, where $e_{c,m}$ represents the embedded pixels close to the m^{th} prototype of the c^{th} class.

(3) Relative Distance Loss: As mentioned earlier, a set of prototypes of class c is defined as $\{\rho_{c,m}\}_{m=1}^M$, where $c \in \{0, 1, \dots, C-1\}$. The predicted classification for each embedded pixel $e \in F_{\text{proj}}$ can be achieved as follows (Zhou et al., 2022):

$$\hat{c}_e = c^*, \text{ with } (c^*, m^*) = \underset{(c,m)}{\text{argmin}} \{ \langle e, \rho_{c,m} \rangle \}_{c,m=1}^{C,M} \quad (7)$$

where $e \in \mathbb{R}^D$ represents the l_2 -normalized embedding vector. The distance measure $\langle \cdot, \cdot \rangle$ represents the negative cosine similarity, i.e., $\langle e, \rho \rangle = -\cos(e, \rho)$, which can also be simplified to $\langle e, \rho \rangle = -e \cdot \rho$. We can use this measure to calculate the distance between the embedded pixel e and the prototype ρ . The class probability distribution of embedded pixel e of class c is defined as (Zhou et al., 2022):

$$p(c|e) = \frac{\exp(-d_{c,e})}{\sum_{c=1}^C \exp(-d_{c,e})}, \text{ with } d_{c,e} = \min \{ \langle e, \rho_{c,m} \rangle \}_{m=1}^M \quad (8)$$

where $d_{c,e}$ represents the distance between the embedded pixel e and its nearest prototype $\rho_{c,m}^c$ for the c^{th} class. We then optimize network training via the cross-entropy loss function using the hard predictions Y^s from the segmentation head of the main branch. This loss pushes the embedded pixel e closer to the nearest prototype of its class and away from other classes, thereby preliminarily optimizing the relative intra-class and inter-class distances (Zhou et al., 2022):

$$\mathcal{L}_{rdl-ce} = - \sum_{c=1}^C Y^s \log p(c|e) \quad (9)$$

where c represents the class, $p(c|e)$ can also be regarded as the similarity map $P^{pro} \in \mathbb{R}^{C \times H \times W}$ obtained by pixel-by-pixel matching between the embedded features and the prototypes.

(4) Absolute Distance Loss: Given that each class is delineated by a set of prototypes, the variations in distribution within a class find expression through these prototypes. Nevertheless, the conventional cross-entropy loss overlooks this association between pixels and prototypes within a class, a concept we term as absolute distance. Consequently, we adopted the absolute distance loss from the reference (Zhou et al., 2022) to enhance both inter-class separation and intra-class compactness. The subsequent section outlines the implementation steps and formulas.

First, the embedded pixel e is pushed closer to its similar prototype (positive sample $\mathcal{P}^+$) and away from dissimilar prototypes (negative samples $\mathcal{P}^-$) through contrastive loss, thereby achieving enhanced inter-class and intra-class discrimination.

$$\mathcal{L}_{adl-cl} = -\log \frac{\exp(e_{c,m} \mathcal{P}^+ / \tau_1)}{\exp(e_{c,m} \mathcal{P}^+ / \tau_1) + \sum_{\mu^- \in \mathcal{P}^-} \exp(e_{c,m} \mu^- / \tau_1)} \quad (10)$$

Absolute distance loss based on contrastive learning ($\mathcal{L}_{adl-cl}$) can push each embedded pixel $e_{c,m}$ close to its similar prototype, i.e., the positive sample (defined as $\mathcal{P}^+ = \mu_{e_c, m_c}$), and away from the dissimilar prototype $\mathcal{P}^- = \{\mu_{e_c, m'}\}_{c,m=1}^{C,M} / \mu_{e_c, m_c}$. Here, $e_{c,m}$ represents the embedded pixels close to the m^{th} prototype of the c^{th} class. The amount of $\mathcal{P}^-$ is $CM - 1$, that is, the whole prototype set excluding $\mathcal{P}^+$. τ_1 is the temperature coefficient, set to 0.1 following (Wang et al., 2021b).

To further optimize the distance between sub-classes within the class, we adopted a subclass compactness loss ($\mathcal{L}_{adl-com}$). This loss minimizes the distance between the embedded pixels $e_{c,m}$ within the class and its corresponding nearest prototype μ_{e_c, m_c} , thereby enhancing the compactness within the class:

$$\mathcal{L}_{adl-com} = \frac{1}{CM} \sum_{c=1}^C \sum_{m=1}^M \|1 - e_{c,m} \mu_{e_c, m_c}\|_2 \quad (11)$$

where C is the number of classes and M is the number of prototypes for each class.

Finally, the overall distance optimization loss can be defined as follows:

$$\mathcal{L}_{dis} = \mathcal{L}_{rdl-ce} + \lambda_1 \mathcal{L}_{adl-cl} + \lambda_2 \mathcal{L}_{adl-com} \quad (12)$$

where the loss function coefficients λ_1 and λ_2 are empirically set to 0.01 and 0.01 respectively (Zhou et al., 2022).

3.4. Boundary regression

After addressing mis-segmentation in OCT retinal layers caused by interference factors, we turned our attention to enhancing boundary learning. Accurate, smooth, and continuous boundaries are essential for the layer segmentation task. In recent deep learning approaches, two common strategies are employed to improve boundaries and mitigate topological errors in boundary regression tasks: model boundary coordinates (He et al., 2021) and distance maps (Liu et al., 2020). However, these strategies typically require densely annotated supervision. In our study, we utilized scribble annotations, which are not fully labeled, posing a challenge for boundary regression tasks. To overcome the lack of dense annotations, we initially employed generated mixed pseudo-labels in lieu of scribble annotations. Subsequently, under the supervision of these pseudo-labels, we enhanced boundary learning and

mitigated topological errors using a combination of 1D coordinate loss and 2D reconstruction loss. The detailed implementation of the boundary regression process is illustrated in Fig. 5, which includes mixed pseudo-label generation and boundary regression loss. Further specifics will be elaborated below.

(1) Mixed pseudo-labels generation: As mentioned earlier, scribble annotations is incomplete and cannot be directly used for supervised boundary regression task. Additionally, using only scribble annotations as supervision limits network performance. To alleviate this problem, we decided to use the output P^{pro} of the uncertainty prototype module as pseudo-labels instead of scribble annotations. However, while the output integrates a good representation of certainty features, it lacks position information. Conversely, the output $T^{-1}(P^t)$ of the CNN-based transform branch is more position-sensitive. Therefore, we mix the output of the uncertainty prototype module with the output of the transform branch to generate mixed pseudo-labels $Y^{mix} \in \mathbb{R}^{H \times W}$ to extend scribble annotations, as shown in Fig. 5(d). In addition, this strategy can also compensate for the weaknesses of the two predictions and avoid the inherent shortcomings of retaining its own prediction without updating when a single prediction is used as a pseudo label (Huo et al., 2021; Luo et al., 2022).

$$Y^{mix} = \text{argmax}(\beta \times P^{pro} + (1 - \beta) \times T^{-1}(P^t)) \quad (13)$$

where $\beta = 0.5$, and the argmax operation is used to generate hard pseudo label.

(2) Boundary Regression Loss: After generating mixed pseudo-labels, we design boundary regression loss to enhance boundary learning and alleviate topological errors. The detailed implementation steps are shown in Fig. 5(d). For an input image $X \in \mathbb{R}^{H \times W}$, the network can generate W independent boundary position distributions $p_j(x_j|X; \theta)$, $j \in \{1, 2, \dots, W\}$ for each A-scan (He et al., 2021), x_j is the row index of j^{th} column for the input image and θ is the network parameter. Therefore, for the output of the regression head from the main branch $P^r \in \mathbb{R}^{B \times H \times W}$, where B is the number of boundaries, H and W are the same size as the input image X , and both represent height and width respectively. We first use cross-entropy loss to train the regression head close to the boundary ground truth:

$$\mathcal{L}_{CE} = - \sum_{j=1}^W \sum_{i=1}^H \mathbb{I}(x_j^g = i) \log p_j(x_j^g | X; \theta) \quad (14)$$

where x_j^g is the boundary position (i.e. row index) of the j^{th} column of ground truth. Note that we use mixed pseudo label Y^{mix} as ground truth. In addition, boundary position distribution $p_j(x_j|X; \theta)$ of column j is obtained by performing a column-wise Gumbel-SoftMax on the output of the regression head P^r , represented as $p_j \in \mathbb{R}^{B \times H}$, $j \in \{1, 2, \dots, W\}$. $\mathbb{I}(x_j^g = r)$ is the indicator function, that is, 1 if the indicator function is true, 0 otherwise.

Then, we employ sampling-argmax (Li et al., 2021) instead of soft-argmax to regress the boundary position. The final boundary position $L \in \mathbb{R}^{B \times W}$ at each column j can be estimated through a weighting operation like soft-argmax:

$$L_{b,j} = \sum_{h=1}^H h \cdot p_{b,h,j} \quad (15)$$

where $L_{b,j}$ and $p_{b,h,j}$ respectively represent the boundary position in the j^{th}

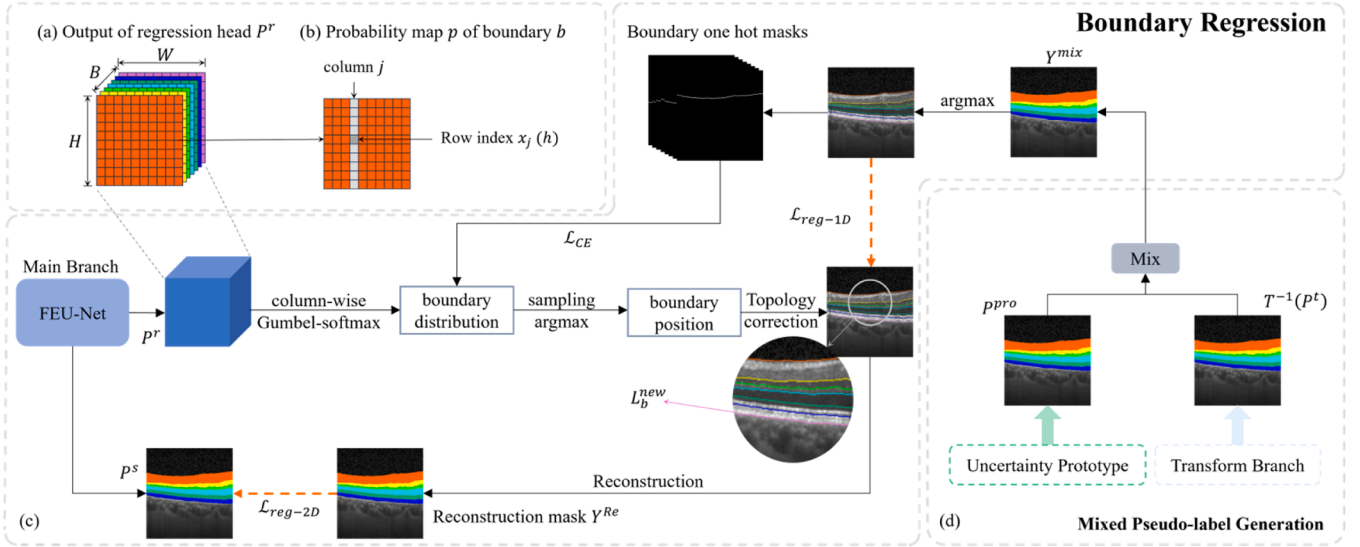


Fig. 5. Illustration of implementation details of boundary regression. (a) Output of the regression head from the main branch. (b) The distribution of vertical coordinates corresponding to each column (i.e., A-scan) of the boundary. (c) Implementation details of boundary regression loss. (d) Mixed pseudo-label generation.

column of the b^{th} boundary and the boundary probability distribution in the j^{th} column of the b^{th} boundary at row h .

To correct the topological order of the retinal layers, we define the topological order of the retinal layer boundary as $L_{1,j} \leq L_{2,j} \leq \dots \leq L_{B,j}$, and then constrain the retina layer through the following formula based on the previous step:

$$L_{b,j}^{new} = L_{b-1,j} + \text{RELU}(L_{b,j} - L_{b-1,j}) \quad (16)$$

where $L_{b,j}$ represents the retinal coordinates on the j^{th} column A-scan. Through the above formula, we try our best to correct the topological order on each A-scan.

Further, Smooth L1 Loss is used to supervise network training to create 1D coordinates L^{new} that maintain topological order closer to ground truth. The 1D coordinate loss ($\mathcal{L}_{reg-1D}$) is

$$\mathcal{L}_{reg-1D} = \sum_{b=1}^B \sum_{j=1}^W \frac{1}{2} d_{b,j}^2 \mathbb{I}(|d_{b,j}| < 1) + \left(|d_{b,j}| - \frac{1}{2}\right) \mathbb{I}(|d_{b,j}| \geq 1) \quad (17)$$

where $d_{b,j} = L_{b,j}^{new} - \text{argmax}(Y_{b,j}^{Mix})$, and the argmax operation is performed in the H dimension.

To further alleviate topological errors, we reconstruct the 2D retinal layer mask $Y^{Re} \in \mathbb{R}^{H \times W}$ using the rectified layer coordinates. Finally, we use the reconstructed mask Y^{Re} to replace mixed pseudo-labels Y^{mix} that may suffer from topological errors and mis-segmentation to supervise the output P^s from the main branch through dice loss ($\mathcal{L}_{dice}$). In this way, the 2D reconstruction loss ($\mathcal{L}_{reg-2D}$) is as follows:

$$\mathcal{L}_{reg-2D} = \mathcal{L}_{dice}(P^s, Y^{Re}) \quad (18)$$

Therefore, the total boundary regression loss is defined as:

$$\mathcal{L}_{reg} = \mathcal{L}_{CE} + \mathcal{L}_{reg-1D} + \lambda_3 \mathcal{L}_{reg-2D} \quad (19)$$

where $\lambda_3 = 0.5$. It is worth noting that the poor prediction of scribble supervision in the initial state is not conducive to the boundary regression task, so we started to add $\mathcal{L}_{reg-2D}$ after warming up for six epochs.

3.5. Loss functions

Fig. 3 illustrates the overview of the proposed scribble-supervised retinal layer segmentation framework. In addition to the losses already described previously, including the distance optimization loss in

the uncertainty prototype module (see details in Section 3.3) and the boundary regression loss from the boundary regression (see details in Section 3.4), we also apply scribble supervised loss, consistency loss, and pseudo-supervised loss to supervise our network training. In this section, we detail these three losses and the total loss for network training. For the input image $X \in \mathbb{R}^{H \times W}$ and the corresponding $T(X)$ of the dual-branch network, the corresponding scribble annotations are $Y \in \mathbb{R}^{H \times W}$ and $T(Y)$, where $T(\cdot)$ is the transformation function. Therefore, we can get the output $P^s \in \mathbb{R}^{C \times H \times W}$ of the segmentation head from the main branch $B_m(\cdot)$, the output $P^t \in \mathbb{R}^{C \times H \times W}$ of the transform branch $B_t(\cdot)$, and the output $P^{pro} \in \mathbb{R}^{C \times H \times W}$ of the uncertainty prototype module. Here, C is set to 8.

(1) Scribble-supervised Segmentation Loss: In this paper, we apply partial cross-entropy loss that ignores unlabeled pixels to scribble supervised segmentation, which can be formulated as:

$$\mathcal{L}_{pce} = -\frac{1}{|\Omega_Y|} \sum_{c=1}^C \sum_{i \in \Omega_Y} Y_{c,i} \log P_{c,i} \quad (20)$$

where $Y_{c,i}$ and $P_{c,i}$ represent the scribble annotation and predicted probability of pixel i in the c^{th} class respectively. Ω_Y is the set of labeled pixels in the scribble annotation. Therefore, the scribble supervision loss ($\mathcal{L}_{ssl}$) of the outputs (i.e., P^s and P^t) from the two branches can be written as:

$$\mathcal{L}_{ssl} = 0.5 \times (\mathcal{L}_{pce}(P^s, Y) + \mathcal{L}_{pce}(P^t, T(Y))) \quad (21)$$

(2) Consistency Loss: To further improve the network's anti-interference ability, we also introduce consistency loss ($\mathcal{L}_{con}$), which minimizes the differences between different predictions through mean square error loss:

$$\mathcal{L}_{con} = 0.5 \times (\mathcal{L}_{mse}(P^s, T^{-1}(P^t)) + \mathcal{L}_{mse}(P^{pro}, T^{-1}(P^t))) \quad (22)$$

where $\mathcal{L}_{mse}$ is the mean square error loss and $T^{-1}(P^t)$ represents the output of the transform branch after inverse transform.

(3) Pseudo-supervised Segmentation Loss: As described in Section 3.4, the mixed pseudo-label Y^{mix} contains more supervision information compared to scribble annotations. Therefore, we use it to supervise the outputs of both branches to further facilitate network training. The pseudo-supervised segmentation loss ($\mathcal{L}_{seg}$) is expressed as:

$$\mathcal{L}_{seg} = 0.5 \times (\mathcal{L}_{dice}(P^s, Y^{mix}) + \mathcal{L}_{dice}(T^{-1}(P^t), Y^{mix})) \quad (23)$$

where $\mathcal{L}_{dice}$ is the dice loss, P^s and $T^{-1}(P^t)$ represent the output of the main branch and the output of the transform branch after inverse transform respectively.

(4) Total Loss: The total loss during the training phase of the proposed W-UPBR is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{sst} + \lambda_{con}\mathcal{L}_{con} + \lambda_{dis}\mathcal{L}_{dis} + \lambda_{seg}\mathcal{L}_{seg} + \lambda_{reg}\mathcal{L}_{reg} \quad (24)$$

where the parameter λ_{con} is set as described in (Tarvainen and Valpola, 2017), employing a time-dependent Gaussian warm-up function $\lambda(t)$ to adjust the weight of the consistency loss, λ_{dis} , λ_{seg} and λ_{reg} are empirically set to 1, 1 and 1, respectively.

4. Experiment results

4.1. Datasets

To evaluate the performance of the proposed W-UPBR, we conducted experiments on three datasets respectively. Two were from eyes with AMD (Cell-AMD (Keremany et al., 2018) and Aier-AMD) and one dataset was with normal eyes (OCTRIMA (Tian et al., 2015)). Table 1 lists the information of the used datasets. Below, we will introduce the basic information of the three datasets and the division of the datasets.

Cell-AMD Dataset (Keremany et al., 2018): Cell-AMD dataset is part of an OCT dataset provided by (Keremany et al., 2018), and the OCT images were acquired using Spectralis SD-OCT (Heidelberg Engineering, Heidelberg, Germany) from multiple hospitals. 150 OCT B-scans were selected from the original dataset and further subdivided into three lesion types, including DRUSEN, PED and CNV. The selected original image size was 496×768 pixels or 512×512 pixels. To ensure the accuracy and authenticity of the scribble annotations, we asked experienced experts to annotate the dataset. These experts classified the OCT B-scan images into eight categories, including two categories of background regions (Vitreous and Choroid) and six categories of layer regions (RNFL+GCIP, INL, OPL, ONL, IS+OS and RPE), where GCIP was

the abbreviation of GCL+IPL. Fig. 2(a) illustrates an example of an OCT B-scan annotation. Then, we randomly selected 105 images as the training set, 15 images as the validation set, and 30 images as the test set. Overall, the scribbled pixels accounted for about 10 % of the full labeled pixels. Usually, the time to manually label a B-scan depends on the number of boundary layers labeled, image quality, and the severity of the lesion. It takes an average of 30 min to label a B-scan (Tian et al., 2015). Using scribble annotations can significantly reduce annotation time, with labeling a single B-scan taking only 6 min on average.

Aier-AMD Dataset: The Aier-AMD dataset was collected using TOPCON DRI SS-OCT scanner device at Wuhan Aier Eye Hospital. It contained 120 OCT images, covering four lesions of two types of AMD, namely DRUSEN, GA, PED and CNV. The original size of each B-scan was 992×1024 pixels. Unlike the Cell-AMD dataset, experienced experts classified the OCT B-scan images into seven categories according to image quality, including Vitreous, RNFL+GCIP, INL+OPL, ONL, IS+OS, RPE and Choroid. Scribble and pixel-level ground truth of all OCT B-scan images were annotated by clinical experts. In our experiments, we randomly selected 84 images for training, 12 images for validation, and 24 images for testing.

OCTRIMA Dataset (Tian et al., 2015): The OCTRIMA dataset was collected using Spectralis SD-OCT (Heidelberg Engineering GmbH, Heidelberg, Germany) and included 10 volume data from 10 healthy adult subjects. Each volume contains 10 B-scans, each with an original size of 496×768 pixels. Seven retinal layers were manually delineated for each B-scan. Including the two background regions, there are a total of nine categories: Vitreous, RNFL, GCIP, INL, OPL, ONL+IS, OS, RPE and Choroid. We used to the previous method (Luo et al., 2022; Valvano et al., 2021) to generate the scribble annotations through an automatic code program. Specifically, we labeled the foreground regions (C – 2 layer regions) and two background regions (Vitreous and Choroid) of each B-scan. To align with the annotators' habits (generally annotating regions far from the boundaries) and considering the structure of the OCT retinal layers, we primarily focused on annotating the central region of the retinal layers, with a small amount of annotation near the upper boundary of each layer. In our experiments, similar to (Liu et al., 2018; Wang et al., 2021a), we randomly selected 80 images for training and 20 images for testing.

4.2. Implementation details

Since there are many upper and lower background regions in an OCT retinal layer image, we cropped each original OCT image from the three datasets used in our experiments. This allowed us to reduce the computational overhead and focus on the region of interest, specifically the retinal layer regions. The cropped sizes are as shown in Table 1. During the training phase, the image size was set to 256×256 through cropping, rotation, and zero padding before being inputted to network for training. The framework of the entire network was built using PyTorch framework, and all experiments were performed on a single NVIDIA GeForce GTX 3090 (24GB). We used the SGD optimizer. The initial learning rate was set to 0.03, the weight decay was $5e-4$, and the batch size was set to 8. The number of prototypes for per class was set to 11 empirically. For detailed details on this parameter, please see the last part of the ablation study. The code is available at: <https://github.com/callmebyZJ/W-UPBR>.

4.3. Comparison methods and metrics

To comprehensively evaluate the performance of the proposed method, we first compared our method with several state-of-the-art scribble-supervised segmentation methods, which included: 1) different scribble-supervised training strategies using only U-Net as the base segmentation network using only partial cross-entropy loss ($UNet_{pce}$) (Tang et al., 2018), using entropy minimization regularization ($UNet_{emr}$) (Grandvalet and Bengio, 2004), and using mumford-shah-Loss

Table 1
Information about the used dataset.

Dataset	Cell-AMD	Aier-AMD	OCTRIMA
Scanner device	Heidelberg Spectralis SD-OCT	TOPCON DRI SS-OCT	Heidelberg Spectralis SD-OCT
Disease	AMD	AMD	Healthy
Original size	496×768 or 512×512	992×1024	496×768
Cropped size	320×512	320×512	256×512
Number of layers	6	5	7
Number of images	150	120	100

($UNet_{msl}$) (Kim and Ye, 2019), 2) other SOTA scribble-supervised method with other architectures, including Scribble2Label (S2L) (Lee and Jeong, 2020), USTM (Liu et al., 2022), DMPLS (Luo et al., 2022), S^2ME (Wang et al., 2023) and ScribFormer (Li et al., 2024). In addition, we also compared our method with three fully supervised methods, U-Net (Ronneberger et al., 2015), ReLayNet (Roy et al., 2017) and nnUNet (Isensee et al., 2021) where the second method specifically focused on the layer segmentation task. It is worth noting that since we are the first to attempt weakly supervised segmentation using scribble annotations on the layer segmentation task, we selected scribble-supervised methods that focus on other segmentation tasks for comparison. In our experiments, the hyperparameters of all compared methods were set according to the original literature.

We use two metrics to quantitatively evaluate the performance of our method W-UPBR: (1) Dice Score is used to measure the overlap between predictions and ground truth, (2) Mean Absolute Distance (MAD) is used to measure the difference between the boundaries of segmentation predictions and the ground truth.

4.4. Results on the Cell-AMD dataset

In this section, we will introduce the qualitative and quantitative comparison results of our proposed W-UPBR method and several comparison methods on the Cell-AMD dataset. Qualitative results are shown in Fig. 6, including Drusen, PED and CNV images. As can be seen from the figure, when dealing with Drusen images (as shown in the first column) with small layer distortion, the overall performance of the listed methods was satisfactory, with no serious topological order errors. Among the compared weak supervision methods, S2L, USTM and S^2ME showed mis-segmentation phenomena, mainly concentrated in the choroid region, as observed in the row (e), (f) and (h) in the first column. In the fully supervised methods, the layer region of the RPE in ReLayNet (row (k)) was also prone to mis-segmentation, with a small part of the choroidal region being incorrectly segmented into the RPE layer region. However, when faced with more severe layer distortions, such as PED (second column) and CNV (third column), the segmentation results of most of the listed methods were unsatisfactory. Not only are there many topological errors, but there are also significant under-segmentation problems in the RPE layer region. For example, when processing B-scans with PED, the segmentation result of $UNet_{msl}$ (as shown in the row (d)) was the worst. In addition to the large gap in the segmentation result of the RPE layer region, there were also under-segmentation or mis-segmentation in other layer regions. When dealing with B-scans with CNV, some weakly supervised methods and fully supervised methods also had the same problems. Moreover, all the compared methods exhibited the phenomenon of over-segmenting the RPE layer region, wherein pixels originally belonging to the IS+OS layer region were classified as part of the RPE layer region. We speculate that this phenomenon is attributed to the local bulge of the RPE and the presence of abnormal blood vessels, both of which impede segmentation. However, the segmentation results of some methods were quite good, such as S^2ME , ScribFormer and nnUNet (see in the row (h), (i) and (l)). For example, ScribFormer only had fewer under-segmentation regions. We believe this occurred because the method integrated local CNN information and global Transformer information to better address severe layer distortion issues. However, it resulted in numerous over-segmentation instances, which can also account for its poor performance, as evidenced by the Dice score and MAD (as indicated in the penultimate row of the methods using scribble labels in Tables 2 and 3, respectively). Although the proposed W-UPBR (row (m)) also exhibited a few of the aforementioned issues, its segmentation results were notably closer to the ground truth (GT) than most methods, as indicated in row (n). We attribute this to the GLCA module and MSF module in our proposed method, which effectively incorporate global and local information and fuse multi-scale features to enhance feature representation,

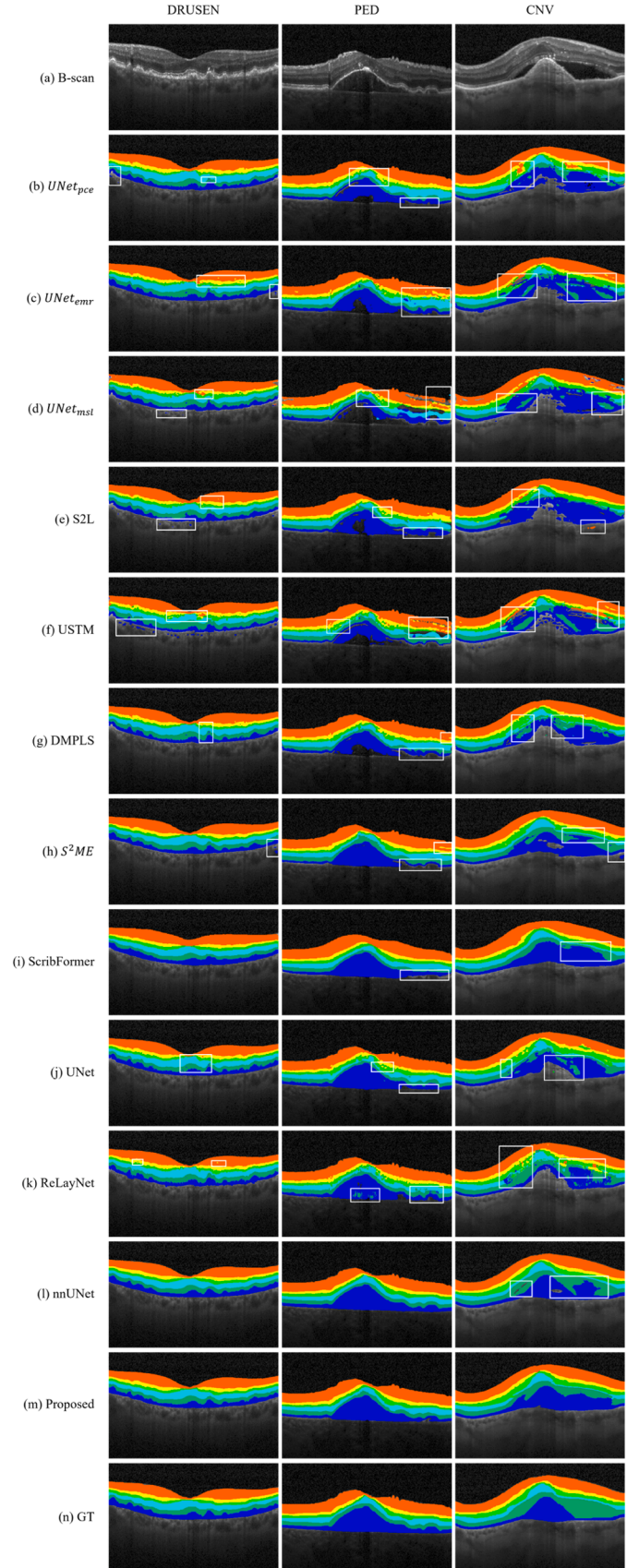


Fig. 6. Segmentation results of B-scan on Cell-AMD data set. We have highlighted some regions with white boxes that are inconsistent with the ground truth (GT). Color mapping of layer regions is explained as follows: ■ RNFL+GCIP, ■ INL, ■ OPL, ■ ONL, ■ IS+OS, ■ RPE.

Table 2

Dice Score $\uparrow$ of each comparison method on the Cell-AMD test dataset of 30 images. Bold is the best in the scribble supervision methods. * and ** represent the p-value of the paired *t*-test between each method and our proposed method W-UPBR (*: $p < 0.05$, **: $p < 0.01$).

Label	Methods	RNFL+GCIP	INL	OPL	ONL	IS+OS	RPE	Overall
Scribble	<i>UNet_{pce}</i> (Tang et al., 2018)	0.923**	0.825**	0.741**	0.876*	0.839**	0.788**	0.832**
	<i>UNet_{emr}</i> (Grandvalet and Bengio, 2004)	0.932**	0.786**	0.733**	0.859**	0.829**	0.792**	0.822**
	<i>UNet_{msl}</i> (Kim and Ye, 2019)	0.904**	0.751**	0.688**	0.822**	0.768**	0.657**	0.765**
	S2L (Lee and Jeong, 2020)	0.931**	0.786**	0.728**	0.833**	0.779**	0.705**	0.794**
	USTM (Liu et al., 2022)	0.920**	0.762**	0.695**	0.822**	0.850*	0.662**	0.785*
	DMPLS (Luo et al., 2022)	0.944**	0.805**	0.747**	0.872*	0.833*	0.773**	0.829**
	<i>S²ME</i> (Wang et al., 2023)	0.949**	0.872	0.831	0.883	0.874	0.830**	0.873*
	ScribFormer (Li et al., 2024)	0.883**	0.780**	0.757**	0.870*	0.799**	0.812**	0.817**
	Proposed (W-UPBR)	0.960	0.879	0.840	0.908	0.900	0.885	0.895
	U-Net (Ronneberger et al., 2015)	0.965*	0.843**	0.800**	0.887	0.872	0.867	0.872*
Full	ReLayNet (Roy et al., 2017)	0.965*	0.854*	0.804**	0.892	0.884	0.838**	0.873*
	nnUNet (Isensee et al., 2021)	0.976**	0.899**	0.867**	0.922**	0.920**	0.919**	0.917**

Table 3

MAD (pixels) $\downarrow$ and standard deviation of each comparison method on the Cell-AMD test dataset of 30 images. Bold is the best in the scribble supervision methods. * and ** represent the p-value of the paired *t*-test between each method and our proposed method W-UPBR (*: $p < 0.05$, **: $p < 0.01$).

Label	Methods	ILM	IPL/INL	INL/OPL	OPL/ONL	ONL/IS	OS/RPE	BM	Overall
Scribble	<i>UNet_{pce}</i> (Tang et al., 2018)	2.405** (4.633)	1.787** (3.811)	1.506* (2.182)	2.345** (4.080)	1.634 (6.075)	3.276 (9.113)	11.120** (33.054)	3.439 (13.841)
	<i>UNet_{emr}</i> (Grandvalet and Bengio, 2004)	1.630** (4.317)	3.279** (14.375)	2.601* (12.312)	2.966** (7.968)	1.990* (5.613)	4.789** (12.921)	10.988** (32.062)	4.035 (15.718)
	<i>UNet_{msl}</i> (Kim and Ye, 2019)	1.671** (3.538)	2.427** (6.360)	3.943* (18.615)	3.482** (8.651)	2.124** (5.389)	8.094** (15.127)	50.268** (63.534)	10.287 (30.848)
	S2L (Lee and Jeong, 2020)	2.128** (5.895)	1.891** (6.814)	2.206** (5.058)	2.636** (4.882)	2.445** (7.043)	5.843** (10.876)	36.181** (69.561)	7.619 (29.520)
	USTM (Liu et al., 2022)	1.544* (3.983)	4.065 (19.568)	4.524 (21.641)	3.520** (11.836)	2.502** (6.739)	6.274** (20.645)	17.472** (37.586)	5.700 (20.931)
	DMPLS (Luo et al., 2022)	1.121 (1.751)	1.889** (4.441)	1.936** (5.651)	2.344* (4.867)	1.986 (11.152)	3.381 (8.262)	7.801** (23.130)	2.923 (10.932)
	<i>S²ME</i> (Wang et al., 2023)	1.900* (4.154)	1.335 (3.310)	1.231 (3.452)	1.961 (5.323)	1.985** (4.520)	2.157 (6.187)	5.578 (19.242)	2.306 (8.541)
	ScribFormer (Li et al., 2024)	4.701** (4.035)	1.866** (2.917)	1.954** (2.283)	2.142 (3.227)	1.814** (5.552)	3.714* (11.069)	3.939 (16.276)	2.876* (8.171)
	Proposed (W-UPBR)	0.972 (1.121)	1.191 (2.643)	1.123 (3.160)	1.617 (4.163)	1.290 (5.594)	1.754 (5.230)	2.114 (7.538)	1.437 (4.660)
	U-Net (Ronneberger et al., 2015)	0.425** (0.677)	1.423 (2.245)	1.512* (2.536)	1.920 (3.019)	1.375 (2.164)	2.227 (5.081)	3.080 (12.590)	1.709 (5.531)
Full	ReLayNet (Roy et al., 2017)	0.447** (1.029)	1.371 (2.344)	1.349 (2.475)	1.802 (3.272)	1.963* (5.851)	2.661 (8.079)	5.721** (20.456)	2.188 (8.934)
	nnUNet (Isensee et al., 2021)	0.314** (0.594)	0.874** (0.892)	0.780** (0.987)	1.318 (1.505)	0.916** (1.101)	1.165* (3.043)	0.989** (0.922)	0.908** (1.527)

enabling better adaptation to severe deformation scenarios. Additionally, our proposed uncertainty prototype module can integrate the similarities and dissimilarities between B-scans, thereby mitigating obvious mis-segmentation in layers caused by interference factors.

Tables 2 and 3 present the quantitative comparison results of Dice score and MAD, respectively. From Table 2, it is evident that the performance of the compared state-of-the-art (SOTA) weakly supervised methods varied significantly under the supervision of scribble annotations. The worst performing method was *UNet_{msl}*, with an average Dice score of only 76.5 %. Under the same scribble supervision, *S²ME* performed better than other SOTA scribble supervision methods, with an average Dice score of 87.3 %, which was comparable to the performance of ReLayNet in the fully supervised methods. In addition, in comparison to the fully supervised method U-Net, the performance of U-Net utilizing scribble annotations, denoted as *UNet_{pce}* (U-Net using partial cross-entropy loss), dropped by 4 % (from 87.2 % to 83.2 %). In comparison, the proposed W-UPBR demonstrated superior performance to all compared methods, achieving an average Dice score of 89.5 %. Not only was it 2.2 % higher than the best-performing scribble-level supervised method *S²ME* (89.5 % vs. 87.3 %), but it also outperformed the two fully supervised methods, U-Net (87.2 %) and ReLayNet (87.3 %), by 2.3 % and 2.2 %, respectively. Furthermore, we conducted a two-tailed paired *t*-test on the results to assess the statistical significance of the differences,

considering $p < 0.05$ as significant. The analysis of the significance on the Dice score indicates that compared to the scribble-supervised method *S²ME*, our method significantly improved the two layer regions of RNFL+GCIP and RPE, with p-values both smaller than 0.01. The significance analysis results, when compared with ReLayNet in the fully supervised methods, also demonstrate that the proposed W-UPBR showed significance in the three layer regions, including INL, OPL, RPE, with p-values of OPL and RPE smaller than 0.01. From Table 3, it is evident that with sparse annotations, the MAD values of the compared SOTA weakly supervised methods all increased (A smaller MAD value signifies a smaller difference between the predicted boundary and the ground truth boundary). Thus, these methods are hindered in generating accurate boundaries due to incomplete annotations. In contrast, our proposed method, W-UPBR exhibited the lowest MAD value across all layer boundaries compared to the SOTA weakly supervised methods. Specifically, our method achieved the lowest average MAD value of 1.437 among the listed methods (except the fully supervised method nnUNet) in Table 3. Remarkably, even when compared with the two fully supervised methods, U-Net (1.709) and ReLayNet (2.188), our W-UPBR showcased a reduction in average MAD value by 0.272 and 0.751, respectively. Significance analysis results on the MAD value underscored the noteworthy improvement achieved by our method compared to most others. We attribute this enhancement to our boundary regression loss

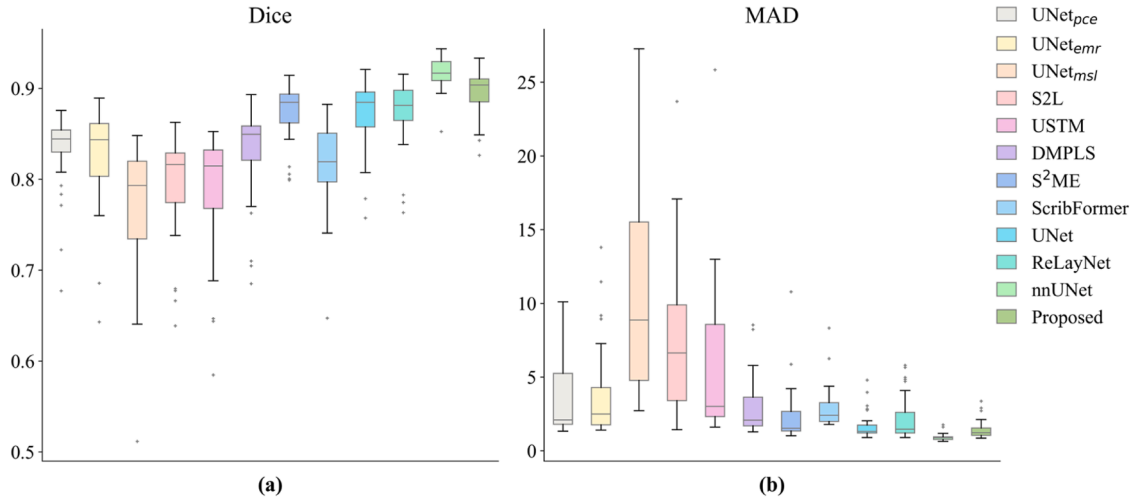


Fig. 7. Boxplots of the Dice Score and MAD comparing the proposed method W-UPBR with other SOTA methods on the Cell-AMD dataset.

mechanism, which minimizes topological errors through 1D coordinate loss. Subsequently, our approach further refined predictions to align closely with ground truth using 2D reconstruction loss, thereby facilitating superior boundary learning.

In addition, we also show the box plots of the quantitative comparison results of Dice score and MAD in Fig. 7. By observing Fig. 7, we can find that our method achieved good results on both evaluation metrics. Specifically, our method had the highest median value on Dice score and the lowest median value on MAD among all compared scribble-supervised methods as well as two fully-supervised methods (UNet and ReLayNet). This indicates that the proposed W-UPBR outperformed the average of most compared methods.

4.5. Ablation study

In this section, we conducted a series of ablation studies to investigate the effectiveness of each component in our model (W-UPBR) on Cell-AMD dataset, including FEU-Net, uncertainty prototype module (UP), pseudo-supervised loss (PSL), boundary regression loss (BRL) and uncertainty guidance.

4.5.1. Effectiveness of FEU-Net

We designed FEU-Net aiming to enhance output features, thereby mitigating the impact of distortions caused by lesions. This architecture incorporates two novel modules: the GLCA module, which facilitates the perception of both global and local information, and the MSF module, designed for fusing multi-scale information. From Table 6, it is evident that, compared to U-Net, FEU-Net improves the average Dice score by 0.7 % with the addition of only 0.22 M parameters added. When employing the dual-branch FEU-Net, the average Dice score improves by 1.1 %, further validating the effectiveness of the dual-branch architecture. Additionally, as shown in Table 4 and Table 5, compared to Model-1 (Dual Branch UNet), Model-2 (Dual Branch FEU-Net) with FEU-Net as

the backbone network improved its average Dice score from 84.8 % to 85.0 %, and the average MAD decreased from 4.399 to 3.827. Among them, IS+OS and RPE, which are easily affected by lesions, increased by 4.5 % (from 80.8 % to 85.3 %) and 1.7 % (from 79.7 % to 81.4 %) in Dice score respectively. Meanwhile, the MAD of the three boundaries involved in these layer regions (i.e., ONL/IS, OS/RPE and BM) was also improved, especially BM decreased by 3.901. This demonstrates that the proposed FEU-Net can effectively cope with the layer distortion by enhancing features.

4.5.2. Effectiveness of uncertainty prototype module

The uncertainty prototype module integrates similarities and dissimilarities among OCT images using multi-category pixel-level prototypes, guided by uncertainty estimation. This integration mitigates false positive predictions in layers attributed to interference factors. Quantitative analysis of the uncertainty prototype module's impact on the method is presented in Tables 4 and 5. Compared to Model-2 (Dual Branch FEU-Net), Model-3 (Dual Branch FEU-Net + UP), which incorporates the uncertainty prototype module based on Model-2, achieved improved segmentation results. The average Dice score increased by 2 % (85.0 % vs. 87.0 %), and the average MAD decreased by 1.480 (3.827 vs. 2.347). More specifically, Model-3 is improved on six boundaries, among which BM is the most significant, and its MAD dropped by 6.498, from 12.017 to 5.519. To qualitatively demonstrate the module's effectiveness, we present experimental results both with and without its inclusion, as depicted in the Fig. 8(a). This illustrates that the proposed uncertainty prototype module effectively mitigated the impact of uncertainty in unlabeled pixels during network training, particularly when scribble supervision lacked complete annotations. Furthermore, it significantly reduced mis-segmentation resulting from interference factors such as noise and artifacts.

Furthermore, scribble annotations can be extended by generating mixed pseudo-labels, thus mitigating the annotation scarcity dilemma.

Table 4
Dice Score of the proposed W-UPBR in the ablation studies on the Cell-AMD dataset.

Models	Retinal layer regions						Overall
	RNFL+GCIP	INL	OPL	ONL	IS+OS	RPE	
(Model-1) Dual Branch UNet	0.946	0.853	0.816	0.870	0.808	0.797	0.848
(Model-2) Dual Branch FEU-Net	0.943	0.860	0.760	0.872	0.853	0.814	0.850
(Model-3) Dual Branch FEU-Net + UP	0.940	0.869	0.801	0.887	0.878	0.845	0.870
(Model-4) Dual Branch FEU-Net + UP + PSL	0.953	0.876	0.836	0.905	0.882	0.861	0.885
(Model-5) W-UPBR w/o Uncertainty Guidance	0.962	0.876	0.835	0.902	0.874	0.850	0.883
(Model-6) W-UPBR (Dual Branch FEU-Net + UP + PSL + BRL)	0.960	0.879	0.840	0.908	0.900	0.885	0.895

Table 5

MAD (pixels) of the proposed W-UPBR in the ablation studies on the Cell-AMD dataset.

Models	Retinal layer boundaries							Overall
	ILM	IPL/INL	INL/OPL	OPL/ONL	ONL/IS	OS/RPE	BM	
(Model-1) Dual Branch UNet	1.346	1.476	1.957	1.895	3.295	4.906	15.918	4.399
(Model-2) Dual Branch FEU-Net	1.685	2.281	2.145	2.948	2.065	3.645	12.017	3.827
(Model-3) Dual Branch FEU-Net + UP	2.137	1.788	1.084	1.984	1.386	2.529	5.519	2.347
(Model-4) Dual Branch FEU-Net + UP + PSL	1.465	1.198	0.955	1.735	1.416	2.681	4.875	2.046
(Model-5) W-UPBR w/o Uncertainty Guidance	0.682	1.266	1.278	1.500	1.356	3.104	4.650	1.977
(Model-6) W-UPBR (Dual Branch FEU-Net + UP + PSL + BRL)	0.972	1.191	1.123	1.617	1.290	1.754	2.114	1.437

Table 6

Dice Score, parameters and computational complexity of FEU-Net and its components on the Cell-AMD dataset.

Method	Retinal layer regions						Overall	Param (M)	FLOPs (G)
	RNFL+GCIP	INL	OPL	ONL	IS+OS	RPE			
U-Net	0.923	0.825	0.741	0.876	0.839	0.788	0.832	1.81	3.04
UNet + GLCA	0.941	0.776	0.751	0.853	0.859	0.834	0.836	1.98	4.07
UNet + GLCA + MSF (FEU-Net)	0.956	0.819	0.782	0.860	0.851	0.767	0.839	2.03	4.57
Dual Branch FEU-Net	0.943	0.860	0.760	0.872	0.853	0.814	0.850	2.03	4.57

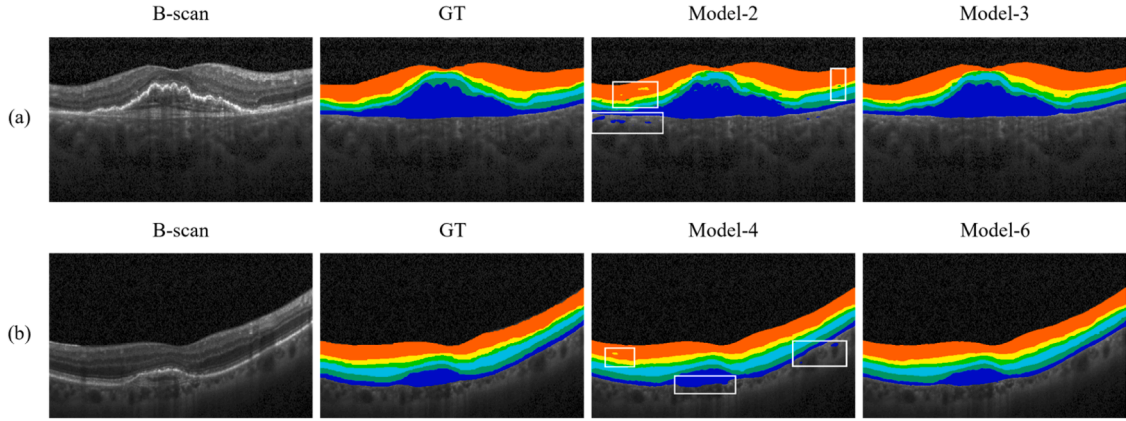


Fig. 8. Illustration of ablation experimental results on the Cell-AMD dataset. (a) Visualization results before and after adding the uncertainty prototype module. (b) Visualization results before and after adding the boundary regression loss. We have highlighted some regions with white boxes that are inconsistent with the ground truth (GT).

As indicated in Tables 4 and 5, under the supervision of high-quality mixed pseudo-labels, Model-4 (Dual Branch FEU-Net + UP + PSL) exhibited improved Dice scores on each layer compared to Model-3 (Dual Branch FEU-Net + UP). Notably, the average Dice score increased by 1.5 % (87.0 % vs. 88.5 %), and the average MAD decreased by 0.301 (2.347 vs. 2.046). This improvement can be attributed to two main factors. Firstly, the ability to extend scribble annotations, as described earlier, played a crucial role. Secondly, the fusion of outputs from two distinct prediction classifiers facilitated the mitigation of inconsistencies between them, thereby accentuating their respective advantages.

4.5.3. Effectiveness of boundary regression loss

The boundary regression loss constrains boundaries in 1D and 2D dimensions, aiming to enhance boundary learning and mitigate topological errors. To evaluate its efficacy, we introduced boundary regression loss in Model-4 (Dual Branch FEU-Net + UP + PSL), resulting in Model-6, i.e., the proposed W-UPBR. The experimental outcomes were detailed in Tables 4 and 5. Compared with Model-4, Model-6 showed improvement in almost every layer region and boundary, with an average Dice score increase of 1 % (88.5 % vs. 89.5 %) and an average MAD decrease of 0.609 (2.046 vs. 1.437). To qualitatively illustrate its

impact, we present segmentation results of Model-4 and Model-6 in Fig. 8(b) to compare outcomes before and after incorporating the boundary regression loss. As depicted in the figure, topological errors were evident prior to the addition of boundary regression loss. Moreover, a slight degree of over-segmentation was observed in the RPE layer region. Conversely, the segmentation outcome of Model-6 closely aligned with the ground truth.

4.5.4. Effectiveness of uncertainty guidance

To mitigate the impact of inaccurate supervision during prototype updates, we introduced uncertainty estimation to generate certainty pseudo-labels. These pseudo-labels are then used to filter out embedded features with high uncertainty, thereby further guiding the prototype updates. The experimental results in Tables 4 and 5 illustrates the significance of uncertainty guidance. Specifically, compared to the network utilizing uncertainty guidance (Model-6, i.e., W-UPBR), in the setting without uncertainty guidance (Model-5, i.e., W-UPBR w/o Uncertainty Guidance), the average Dice score dropped by 1.2 % (89.5 % vs. 88.3 %) and the MAD increased by 0.54 (1.437 vs. 1.977). This underscored the effectiveness of the strategy employing uncertainty guidance in improving network performance.

Table 7

Dice Score and MAD (pixels) of different prototype numbers for per class on the Cell-AMD dataset.

Number	1	3	5	7	9	11	13	15
Dice	0.893	0.892	0.893	0.890	0.892	0.895	0.893	0.891
MAD	1.579 (6.158)	1.559 (4.845)	1.458 (4.446)	1.497 (4.548)	1.474 (3.669)	1.437 (4.660)	1.856 (8.834)	1.535 (5.439)

4.5.5. Impact of prototype number

Table 7 illustrates the influence of varying prototype numbers per class on network performance. It is evident from the table that the best performance was attained when the number of prototypes per class was set to 11. This configuration yielded the highest average Dice score of 89.5 % and the lowest average MAD value of 1.437. Consequently, in our experiments, we set the number of prototypes for each class to 11.

4.5.6. Hyperparameter setting analysis

To explore the impact of the number of forward propagation times N in uncertainty estimation and the weight of the loss function on model performance, we conducted a series of ablation experiments to identify the effects for these hyperparameters. First, we examined the effect of the number of uncertainty estimation forward propagation times on model performance, with the experimental results are shown in Table 8. As shown in the table, while the model performs is best when $N = 16$, considering the balance between computational efficiency and model performance, N is set to 8. Secondly, since there are many loss functions in this paper, we combine our experience and refer to previous similar work to determine the weights of these loss functions, where λ_{con} uses the aforementioned $\lambda(t)$, while λ_{dis} is assigned two values: 0.5 and 1. Given that λ_{pse} and λ_{reg} are supervised by pseudo-labels, we set their weight values to be same and assign three different values: 0.5, 1 and $\lambda(t)$. The experimental results are shown in Table 9. When $\lambda_{dis} = 1$, $\lambda_{pse} = 1$ and $\lambda_{reg} = 1$, the model performance is better.

4.6. Results on the Aier-AMD dataset

To further evaluate the performance of our proposed W-UPBR, we conducted tests on the Aier-AMD dataset. Initially, we present qualitative comparison results between our proposed W-UPBR and other methods, as depicted in Fig. 9. Upon examination of rows (e) and (f) in Fig. 9, it becomes apparent that S2L and USTM methods, which utilize Exponential Moving Average (EMA), exhibited poor performance when processing PED and CNV B-scan images. Notably, the mis-segmentation region within the RPE layer was extensive, accompanied by significant topological inconsistencies. We speculate that the utilization of EMA may lead to the gradual accumulation of erroneous pixels, thereby amplifying the presence of erroneous regions in the final segmentation output. In addition, there are three weakly supervised methods (DMPLS, S^2ME and ScribFormer) like our method, whose segmentation results are shown in row (g), row (h) and row (i) of Fig. 9. All these methods further employed pseudo-supervision by generating mixed pseudo-labels to optimize network training. It is evident that the segmentation results of the latter two methods were superior, exhibiting fewer topological inconsistencies issues. Similar to the Cell-AMD dataset, regions where these problems arose were primarily concentrated in the RPE layer region. In contrast, as illustrated in row (m) of Fig. 9, the segmentation outcomes of our proposed method, W-UPBR, closely resembled the

Table 8Dice Score and MAD (pixels) for different forward propagation times N for uncertainty estimation on the Cell-AMD dataset.

N	1	2	4	8	16
Dice	0.891	0.892	0.893	0.895	0.895
MAD	1.511 (5.464)	1.686 (7.835)	1.590 (6.849)	1.437 (4.660)	1.333 (3.033)

Table 9

Dice Score and MAD (pixels) for different loss function weights on the Cell-AMD dataset.

λ_{dis}	λ_{pse}	λ_{reg}	Dice	MAD
0.5	0.5	0.5	0.892	1.471 (5.293)
0.5	1	1	0.895	1.714 (8.551)
0.5	$\lambda(t)$	$\lambda(t)$	0.875	2.372 (11.429)
1	0.5	0.5	0.891	1.394 (3.094)
1	1	1	0.895	1.437 (4.660)
1	$\lambda(t)$	$\lambda(t)$	0.875	2.604 (1.176)

ground truth (as depicted in row (n)) compared to most methods, with only minor instances of mis-segmentation. Additionally, when handling B-scans with Drusen and GA, most methods demonstrated improved overall segmentation result. Among them, when handling of B-scans with GA, the four methods, S2L, S^2ME , ScribFormer and nnUNet exhibited better. This is evident in rows (e), (h), (i) and (l) of the third column, respectively, where their segmentation results closely align with the ground truth. Regarding other methods, they may misclassify a small number of choroidal regions as the RPE layer region, leading to a minor mis-segmentation of pixels. However, in the provided figure, the segmentation results of our proposed W-UPBR (observed in the row (m)) lacked such mis-segmented pixels and was more closely aligned with the ground truth. We attribute this improvement to our uncertainty prototype module, which effectively integrated both similarities and dissimilarities between OCT images, thereby reducing the occurrence of mis-segmentation across layers.

Tables 10 and 11 list our quantitative evaluation results of Dice score and MAD value on the Aier-AMD dataset. It can be seen from Tables 10 to 11 that the proposed W-UPBR demonstrated strong performance in both average Dice score and MAD value. Compared with the best SOTA weakly supervised method ScribFormer, the average Dice score of W-UPBR increased by 2.2 %, while the MAD value decreased by 0.829. Furthermore, when compared to the fully supervised method U-Net, the proposed W-UPBR achieved a higher average Dice score by 1.2 % and a lower average MAD value by 0.741. The significance analysis, as presented in Tables 10 and 11, demonstrates a notable improvement with our method, W-UPBR. Particularly, significant enhancement was indicated in the layer region of RPE and the layer boundaries (OS/RPE and BM), which may be attributed to the ability of the designed FEU-Net to mitigating the impact of distortions caused by lesions.

4.7. Results on the OCTRIMA dataset

We investigated the generalization performance of the proposed W-UPBR on OCTRIMA, a publicly available dataset comprising healthy eyes. Qualitative results are depicted in Fig. 10. It is evident from the figure that among the compared weakly supervised methods, several of them exhibited topological errors, primarily concentrated in the choroidal region such as $UNet_{pce}$, $UNet_{emr}$ and $UNet_{msl}$ (as illustrated in Fig. 10(b), (c) and (d)). We attribute this discrepancy to the interference from choroidal blood vessels. Nevertheless, there were several methods demonstrating satisfactory performance. For instance, ScribFormer (as shown in Fig. 10(i)) and the proposed W-UPBR (as shown in the Fig. 10 (m)) closely resembled the ground truth. However, despite this visual closeness, the Dice score and MAD value of ScribFormer in Tables 12 and 13 do not reflect its performance favorably. We speculate that although

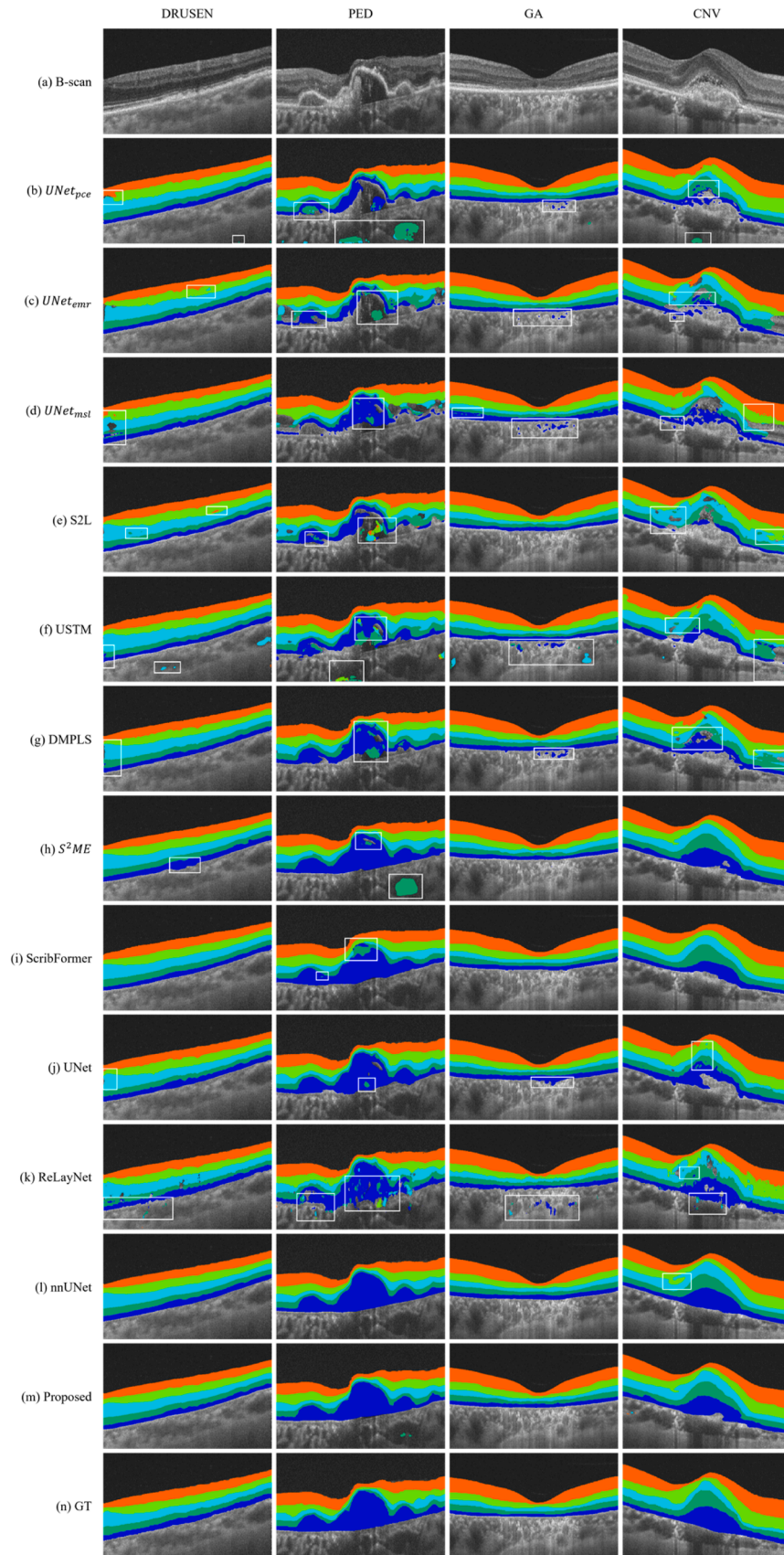


Fig. 9. Segmentation results of B-scan on Aier-AMD dataset. We have highlighted some regions with white boxes that are inconsistent with the ground truth (GT). Color mapping of layer regions is explained as follows: ■ RNFL+GCIP, ■ INL+OPL, ■ ONL, ■ IS+OS, ■ RPE.

Table 10

Dice Score $\uparrow$ of each comparison method on the Aier-AMD test dataset of 24 images. Bold is the best in the scribble supervision methods. * and ** represent the p-value of the paired t -test between each method and our proposed method W-UPBR (*: $p < 0.05$, **: $p < 0.01$).

Label	Methods	RNFL+GCIP	INL+OPL	ONL	IS+OS	RPE	Overall
Scribble	<i>UNet_{pce}</i> (Tang et al., 2018)	0.915**	0.838	0.775*	0.815**	0.755**	0.820*
	<i>UNet_{emr}</i> (Grandvalet and Bengio, 2004)	0.911**	0.736**	0.681**	0.793**	0.739**	0.772**
	<i>UNet_{msl}</i> (Kim and Ye, 2019)	0.897**	0.804**	0.700**	0.692**	0.697**	0.758*
	S2L (Lee and Jeong, 2020)	0.905**	0.746**	0.685**	0.771**	0.686**	0.759**
	USTM (Liu et al., 2022)	0.893**	0.739**	0.696**	0.775**	0.716**	0.764**
	DMPLS (Luo et al., 2022)	0.916**	0.778**	0.720**	0.811**	0.804**	0.806**
	<i>S²ME</i> (Wang et al., 2023)	0.927**	0.836	0.779**	0.822**	0.829**	0.839*
	ScribFormer (Li et al., 2024)	0.884**	0.856	0.795	0.842	0.824**	0.840
	Proposed (W-UPBR)	0.937	0.849	0.812	0.863	0.851	0.862
	U-Net (Ronneberger et al., 2015)	0.953**	0.854	0.770*	0.840*	0.830*	0.850
Full	ReLayNet (Roy et al., 2017)	0.937	0.828	0.772**	0.823**	0.823*	0.837*
	nnUNet (Isensee et al., 2021)	0.959**	0.888**	0.839*	0.889**	0.871**	0.889**

Table 11

MAD (pixels) $\downarrow$ and standard deviation of each comparison method on the Aier-AMD test dataset of 24 images. Bold is the best in the scribble supervision methods. * and ** represent the p-value of the paired t -test between each method and our proposed method W-UPBR (*: $p < 0.05$, **: $p < 0.01$).

Label	Methods	ILM	IPL/INL	OPL/ONL	ONL/IS	OS/RPE	BM	Overall
Scribble	<i>UNet_{pce}</i> (Tang et al., 2018)	1.223** (1.191)	4.727** (7.177)	4.553 (8.469)	3.075* (4.111)	2.890** (6.880)	8.706** (19.695)	4.196 (10.077)
	<i>UNet_{emr}</i> (Grandvalet and Bengio, 2004)	1.449** (1.039)	4.538** (7.847)	9.128** (16.081)	5.446** (10.182)	2.803 (6.203)	16.395** (29.858)	6.626 (15.831)
	<i>UNet_{msl}</i> (Kim and Ye, 2019)	1.493** (1.505)	5.212** (6.460)	5.542* (8.427)	4.746** (8.721)	6.677** (13.270)	20.411** (33.675)	7.347 (16.937)
	S2L (Lee and Jeong, 2020)	1.794** (1.362)	4.430** (4.833)	7.802** (8.580)	6.139** (9.431)	3.692** (10.666)	15.165** (26.968)	6.503 (13.785)
	USTM (Liu et al., 2022)	2.861** (1.582)	4.229* (4.263)	7.948** (7.884)	4.899** (7.794)	4.131* (11.284)	10.000** (18.894)	5.678* (10.526)
	DMPLS (Luo et al., 2022)	1.699** (1.394)	3.835* (5.075)	7.416** (12.344)	4.767** (7.466)	3.632 (12.545)	9.661* (22.144)	5.168* (12.417)
	<i>S²ME</i> (Wang et al., 2023)	1.582** (2.317)	3.346 (2.740)	4.321 (4.378)	3.430** (5.212)	2.618** (2.325)	4.017 (9.221)	3.219** (5.076)
	ScribFormer (Li et al., 2024)	4.468** (2.796)	3.551 (2.931)	3.863 (4.089)	2.861 (2.961)	2.544* (3.043)	3.622 (4.189)	3.485 (3.443)
	Proposed (W-UPBR)	0.801 (1.056)	3.303 (2.645)	3.985 (4.143)	2.712 (3.215)	2.012 (2.571)	3.124 (3.498)	2.656 (3.181)
	U-Net (Ronneberger et al., 2015)	0.382** (0.680)	2.820 (3.076)	4.477 (5.549)	3.645* (4.968)	3.229 (9.806)	5.831** (13.108)	3.397 (7.637)
Full	ReLayNet (Roy et al., 2017)	0.363** (0.610)	3.898 (5.412)	5.015 (7.223)	5.194** (13.904)	3.103** (8.856)	8.005** (16.274)	4.263 (10.413)
	nnUNet (Isensee et al., 2021)	0.292** (0.544)	2.534* (2.755)	3.038** (3.696)	2.318 (2.405)	1.653 (2.286)	2.762 (2.816)	2.099** (2.754)

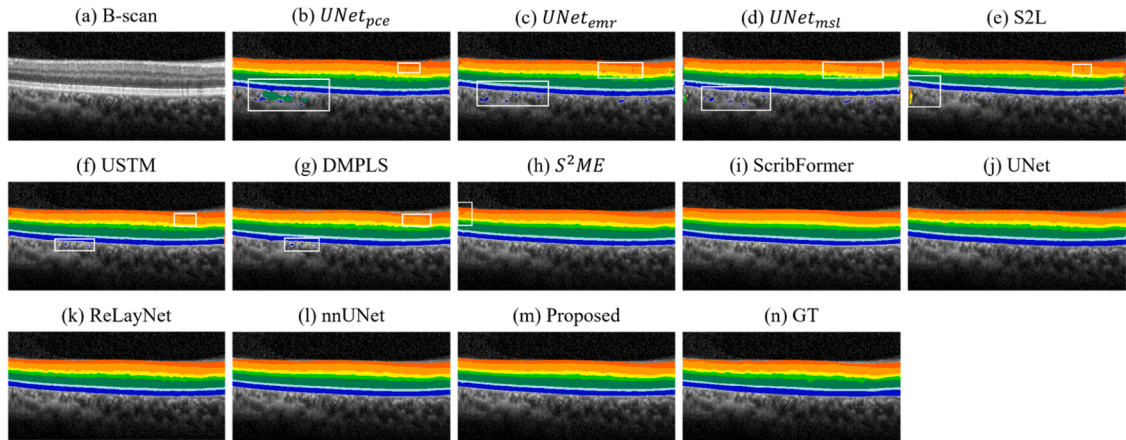


Fig. 10. Segmentation results of B-scan on OCTRIMA data set. We have highlighted some regions with white boxes that are inconsistent with the ground truth (GT). Color mapping of layer regions is explained as follows: ■ RNFL, ■ GCIP, ■ INL, ■ OPL, ■ ONL+IS, ■ OS, ■ RPE.

this method can perceive global information, it may lack detailed processing, resulting in boundary deviations, as evidenced by its higher MAD value. In contrast, our proposed W-UPBR effectively addressed this

issue under the supervision of boundary regression loss.

For quantitative comparison, the evaluation results of two metrics (Dice score and MAD) of all listed methods are presented in [Tables 12](#)

Table 12

Dice Score $\uparrow$ of each comparison method on the OCTRIMA test dataset of 20 images. Bold is the best in the scribble supervision methods. * and ** represent the p-value of the paired t -test between each method and our proposed method W-UPBR (*: $p < 0.05$, **: $p < 0.01$).

Label	Methods	RNFL	GCIP	INL	OPL	ONL+IS	OS	RPE	Overall
Scribble	<i>UNet_{pce}</i> (Tang et al., 2018)	0.830**	0.927*	0.861*	0.842*	0.947*	0.893	0.886**	0.884**
	<i>UNet_{emr}</i> (Grandvalet and Bengio, 2004)	0.868	0.918**	0.836**	0.826**	0.947*	0.891	0.902**	0.884**
	<i>UNet_{msl}</i> (Kim and Ye, 2019)	0.852*	0.922**	0.833**	0.823**	0.947**	0.888	0.879**	0.877**
	S2L (Lee and Jeong, 2020)	0.851*	0.925**	0.845**	0.816**	0.946**	0.895	0.905*	0.883**
	USTM (Liu et al., 2022)	0.858	0.925**	0.847**	0.842*	0.950*	0.858**	0.881**	0.880**
	DMPLS (Luo et al., 2022)	0.867	0.929**	0.841**	0.832**	0.949*	0.889	0.913	0.888**
	<i>S²ME</i> (Wang et al., 2023)	0.854	0.944	0.871	0.844	0.958	0.911	0.927	0.901*
	ScribFormer (Li et al., 2024)	0.738**	0.920**	0.825**	0.837*	0.955	0.897	0.916	0.870
	Proposed (W-UPBR)	0.888	0.949	0.884	0.866	0.964	0.912	0.932	0.914
	U-Net (Ronneberger et al., 2015)	0.910	0.950	0.881	0.871	0.964	0.915	0.940	0.919
Full	ReLayNet (Roy et al., 2017)	0.906	0.949	0.876	0.861	0.963	0.919	0.939	0.916
	nnUNet (Isensee et al., 2021)	0.920**	0.955**	0.892**	0.879**	0.968**	0.927**	0.948**	0.927**

Table 13

MAD (pixels) $\downarrow$ and standard deviation of each comparison method on the OCTRIMA test dataset of 20 images. Bold is the best in the scribble supervision methods. * and ** represent the p-value of the paired t -test between each method and our proposed method W-UPBR (*: $p < 0.05$, **: $p < 0.01$).

Label	Methods	ILM	RNFL/GCL	IPL/INL	INL/OPL	OPL/ONL	IS/OS	OS/RPE	BM	Overall
Scribble	<i>UNet_{pce}</i> (Tang et al., 2018)	2.623* (10.508)	1.399* (2.523)	1.121* (2.208)	1.104* (2.361)	1.427 (4.595)	0.573* (2.500)	0.960* (3.806)	2.742 (11.639)	1.494* (6.213)
	<i>UNet_{emr}</i> (Grandvalet and Bengio, 2004)	1.212 (4.504)	1.509** (6.659)	1.872** (5.634)	1.541** (6.507)	1.784** (6.926)	0.860** (5.841)	0.963 (4.770)	3.416 (14.288)	1.645** (7.518)
	<i>UNet_{msl}</i> (Kim and Ye, 2019)	1.453 (4.322)	1.544** (6.987)	1.959** (7.656)	1.767** (6.488)	2.026** (7.570)	2.273** (10.773)	1.238** (6.825)	3.276 (13.862)	1.942** (7.976)
	S2L (Lee and Jeong, 2020)	1.340 (2.048)	1.362** (5.465)	1.539** (4.439)	1.366** (5.710)	1.930** (6.703)	0.697** (4.963)	0.867 (2.476)	2.543 (13.300)	1.456** (6.532)
	USTM (Liu et al., 2022)	1.199 (2.990)	1.300** (4.614)	1.518** (4.615)	1.245** (4.936)	1.638* (6.611)	0.738** (4.011)	1.156** (1.748)	3.069 (12.271)	1.483** (6.049)
	DMPLS (Luo et al., 2022)	1.197 (2.174)	1.053 (1.448)	1.361** (1.202)	1.143** (1.888)	1.418 (2.452)	0.679** (1.862)	0.803 (0.879)	3.690 (19.581)	1.418 (7.175)
	<i>S²ME</i> (Wang et al., 2023)	1.650** (1.411)	0.927 (2.333)	1.019 (0.966)	1.018 (1.553)	1.287 (1.396)	0.439 (0.532)	0.690 (0.770)	0.831 (0.724)	0.983 (1.373)
	ScribFormer (Li et al., 2024)	3.581** (2.599)	1.288** (1.210)	1.509** (1.232)	1.254** (1.055)	1.233 (1.254)	0.563 (1.067)	0.784 (0.849)	0.959 (0.864)	1.396 (1.625)
	Proposed (W-UPBR)	1.120 (1.659)	0.874 (1.276)	0.891 (0.862)	0.905 (0.859)	1.067 (1.289)	0.431 (0.637)	0.670 (0.770)	1.887 (10.719)	0.981 (3.948)
	U-Net (Ronneberger et al., 2015)	0.727* (1.326)	0.788 (1.002)	0.908 (0.888)	0.897 (0.846)	1.036 (1.232)	0.448 (1.853)	0.665 (0.751)	0.676* (1.611)	0.768 (1.256)
Full	ReLayNet (Roy et al., 2017)	0.703* (1.080)	0.881 (1.358)	0.909 (1.006)	0.975 (0.899)	1.102 (1.254)	0.396 (0.510)	0.755 (2.583)	3.482 (19.460)	1.150 (7.058)
	nnUNet (Isensee et al., 2021)	0.670** (1.283)	0.685** (0.813)	0.796** (0.765)	0.841 (0.807)	0.933* (1.109)	0.352* (0.486)	0.557* (0.663)	0.564* (0.619)	0.675 (0.871)

and 13, respectively. Notably, the experimental results achieved on this dataset exhibited significant improvements relative to those obtained on the two previously described datasets with AMD. We observed that the proposed W-UPBR exhibited commendable performance compared to the majority of the SOTA weakly supervised methods. When compared with the top-performing scribble supervision method *S²ME*, the proposed W-UPBR exhibited notable enhancements. W-UPBR achieved an average Dice score improvement of 1.3 %, rising from 90.1 % to 91.4 %. Additionally, the average MAD value decreased marginally by 0.002, from 0.983 to 0.981. When compared with the two fully supervised methods (UNet and ReLayNet), our proposed method significantly closed the performance gap. Especially, in the two layer regions of INL (88.4 % vs. 88.1 %) and ONL+IS (96.4 % vs. 96.4 %), the Dice scores of our method were either comparable to or even higher than those of U-Net. The paired t -test results indicated that the improvement brought by our method was significant across most methods.

4.7. Discussion

4.7.1. Analysis of model generalizability

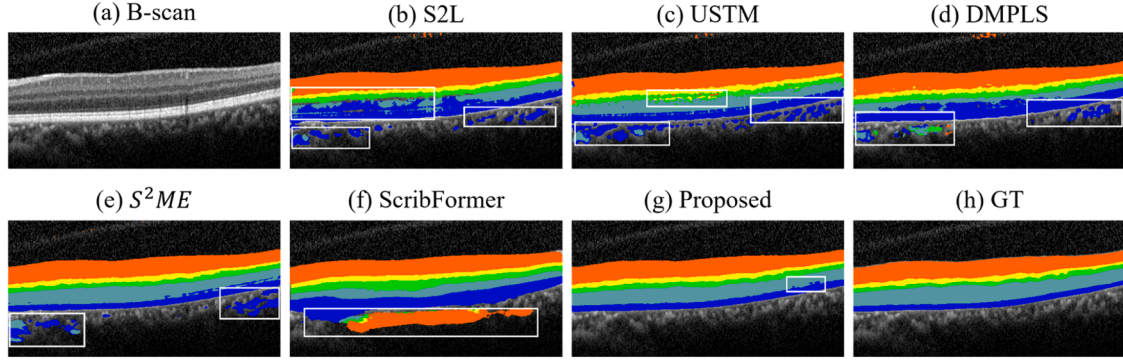
To evaluate the generalizability of the proposed W-UPBR, we conducted two experiments to assess the performance of cross-dataset evaluation according to the retinal layers categorized by the datasets:

1) trained on the Aier-AMD dataset and evaluated on the Cell-AMD dataset, while 2) trained on the Cell-AMD dataset and evaluated on the OCTRIMA dataset. The experimental results are shown in Table 14 and Fig. 11, respectively.

For Set-1, we chose to train on the Aier-AMD dataset and evaluate on the Cell-AMD test set. This is because, compared to the Aier-AMD dataset, the Cell-AMD dataset splits INL and OPL as two independent layers. Therefore, during quantitative evaluation, we can align the predicted categories by merging INL and OPL in the GTs of the Cell-AMD test set. From the Table 14, it is evident that the proposed W-UPBR demonstrates strong robustness, achieving the highest average Dice score of 68.4 % among the compared scribble-supervised methods. It is worth noting that these two datasets were collected using different devices: the Aier-AMD dataset was obtained from SS-OCT, while the Cell-AMD dataset was collected from SD-OCT. The results indicate that, compared to other methods, the proposed W-UPBR also has better generalization performance on unseen cross-domain datasets. In addition, in Set-2, we present the qualitative evaluation results of training on the Cell-AMD dataset and testing on the OCTRIMA test set, as illustrated in the Fig. 11. It is important to note that, to ensure rationality and maintain consistency in the GT of the two datasets, we merged the ONL and IS+OS of the Cell-AMD dataset into one layer region (i.e., ONL+IS+OS). Similarly, we merged the RNFL and GCIP in the

Table 14Dice Score $\uparrow$ of compared methods trained on the Aier dataset and tested on the Cell-AMD dataset. Bold is the best.

Label	Methods	RNFL+GCIP	INL+OPL	ONL	IS+OS	RPE	Overall
Scribble	S2L	0.718	0.423	0.237	0.257	0.262	0.379
	USTM	0.498	0.439	0.221	0.299	0.432	0.378
	DMPLS	0.771	0.557	0.626	0.521	0.446	0.584
	S^2ME	0.873	0.798	0.774	0.332	0.567	0.669
	ScribFormer	0.439	0.316	0.310	0.284	0.232	0.316
	Proposed (W-UPBR)	0.883	0.745	0.778	0.622	0.393	0.684

**Fig. 11.** Segmentation results trained on the Cell-AMD dataset and tested on the OCTRIMA dataset. We have highlighted some regions with white boxes that are inconsistent with the ground truth (GT). Color mapping of layer regions is explained as follows: ■ RNFL+GCIP, ■ INL, ■ OPL, ■ ONL+IS+OS, ■ RPE.

OCTRIMA dataset into one layer region (i.e., RNFL+GCIP), and merged ONL+IS and OS into one layer region (i.e., ONL+IS+OS) when displaying the segmentation results. It can be observed that the segmentation results of the proposed W-UPBR are closer to GT, with fewer mis-segmentation results. The cross-dataset validation results for the above two settings indicate that the proposed W-UPBR demonstrates a higher generalization ability compared to the other scribble-supervised methods.

4.7.2. Analysis of computational complexity

Table 15 shows the parameters, FLOPs, and inference time of our method compared with other deep learning methods. In terms of parameter amount, our method ranks third, only 0.51 M higher than the method based on U-Net. Compared with ScribFormer, the proposed W-UPBR model has 20 times fewer parameters and better performance. In addition, the model complexity of our method is relatively low, ranking second with only 4.57G. With such relatively low parameter amount and complexity, the inference time of a single image of W-UPBR is 18.83ms. Despite the increase in inference time, the segmentation performance is better than other methods except the fully supervised nnUNet. In terms of clinical implications, our method significantly reduces the annotation time without losing accuracy.

Table 15Illustration of model parameters, computational complexity, and inference time for a single 256×256 image.

Method	Parameters (M)	FLOPs (G)	Inference Time (ms)
$UNet_{pce}$	1.81	3.04	4.53
$UNet_{emr}$	1.81	3.04	4.53
$UNet_{msl}$	1.81	3.04	4.53
S2L	1.81	3.04	4.53
USTM	3.62	3.04	4.53
DMPLS	2.45	3.04	4.53
S^2ME	8.74	3.04	4.53
ScribFormer	47.93	44.81	89.74
U-Net	1.81	3.04	4.53
ReLayNet	0.78	16.82	3.74
nnUNet	40.81	16.37	27.3
W-UPBR	2.32	4.57	18.83

4.7.3. Study on the sensitivity of scribble annotations

In practice, the quality of scribble annotation often varies depending on the annotator's techniques and preferences. Discrepancies in scribble annotations among different annotators can significantly impact approaches relying on such supervision. Moreover, the quantity of images also plays a crucial role in determining network performance. Therefore, in this study, we conducted two sets of experiments on the Cell-AMD dataset to examine the influence of both the quality and quantity of scribble annotations on the performance of the proposed W-UPBR. Specifically, 1) Setting 1 involved reducing scribble annotations on a single B-scan, while 2) Setting 2 entailed reducing the number of B-scans.

Table 16 presents the Dice scores of our experimental results under two settings. We observed that as the number of scribble annotations decreased, our network's performance remained relatively stable. In Setting 1, with a reduction from 100 % to 50 % scribble annotations per B-scan, the average Dice score only decreased by 1.1 %, from 89.5 % to 88.4 %. In Setting 2, we halve the number of B-scans only led to 1.3 %, from 89.5 % to 88.2 %. However, at a 60 % reduction, the average Dice score remained unchanged. This can be attributed to the elimination of erroneous annotations in the dataset at this ratio, thereby mitigating their impact on model performance. These experiments in both settings confirmed the stability of the proposed W-UPBR in the face of changes in both the quality and quantity of scribble annotations.

5. Conclusion

In this study, we introduced a novel approach for OCT retinal layer segmentation employing a dual-branch network architecture, incorporating an uncertainty prototype module and a boundary regression loss, utilizing scribble annotations for network supervision. The backbone network of this architecture, termed the Feature Enhancement U-Net (FEU-Net), extends the basic U-Net with two specially designed modules: the Global-Local Context-Aware (GLCA) module for extracting both global and local information to address pathological distortions, and the Multi-Scale Fusion (MSF) module for integrating multi-scale features to enhance and enrich representations. Subsequently, we integrate OCT image similarities and dissimilarities through an uncertainty prototype

Table 16

Dice Score for scribble annotation sensitivity experiments in two settings on the Cell-AMD dataset.

Setting	Ratio	RNFL+GCIP	INL	OPL	ONL	IS+OS	RPE	Overall
Set-1	100 %	0.960	0.879	0.840	0.908	0.900	0.885	0.895
	90 %	0.959	0.876	0.837	0.909	0.900	0.870	0.892
	80 %	0.961	0.876	0.843	0.907	0.893	0.868	0.891
	70 %	0.957	0.872	0.837	0.909	0.896	0.864	0.889
	60 %	0.964	0.876	0.841	0.906	0.890	0.845	0.887
	50 %	0.954	0.870	0.836	0.906	0.885	0.855	0.884
Set-2	100 %	0.960	0.879	0.840	0.908	0.900	0.885	0.895
	90 %	0.961	0.872	0.841	0.909	0.894	0.877	0.892
	80 %	0.963	0.875	0.840	0.903	0.888	0.858	0.888
	70 %	0.959	0.876	0.838	0.908	0.878	0.855	0.886
	60 %	0.960	0.874	0.839	0.905	0.879	0.857	0.886
	50 %	0.961	0.869	0.839	0.902	0.883	0.838	0.882

module to mitigate mis-segmentation and improve accuracy. Furthermore, we employ pseudo-labels generated by combining the outputs of uncertainty prototype module and the transform branch to supervise boundary regression, aiding in mitigating the lack of annotations for scribble labels. This approach effectively reduces retinal layer topology errors through 1D coordinate loss and 2D reconstruction loss. Experimental validation on three datasets demonstrates the rationale and efficacy of our method. However, it should be noted that our method operates solely on 2D slice (single B-scans), thereby failing to exploit anatomical information across adjacent slices in OCT volume (Liu et al., 2024). This limitation may impact segmentation accuracy. Future research directions include exploring methodologies for processing OCT 3D volume data and devising more effective label annotation strategies to harness volumetric information fully for retinal layer segmentation. Additionally, efforts will be directed towards enhancing model accuracy in handling OCT images with severe lesions to better align with clinical requirements.

CRediT authorship contribution statement

Xiaoming Liu: Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Jia Zhang:** Writing – review & editing, Writing – original draft, Software, Methodology. **Ying Zhang:** Writing – review & editing, Resources, Data curation. **Li Chen:** Writing – review & editing, Methodology. **Liangfu Luo:** Writing – review & editing, Resources. **Jinshan Tang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grants 62176190 and 62271359.

Appendix

The procedure of the proposed W-UPBR is listed in Algorithm 1. During the training phase, the algorithm takes the OCT B-scan image X_i and its corresponding scribble annotation Y_i as the input source and transforms the input into $T(X_i)$ and $T(Y_i)$ through the transform operation $T(\cdot)$. Then, X_i is input into the main branch $B_m(\cdot)$ to obtain the output P_i^s of the segmentation head, the output P_i^r of the regression head and the certainty pseudo-label Y_i^u . Meanwhile, $T(X_i)$ is fed into the transform branch $B_t(\cdot)$ to obtain the output P_i^t . Next, the features F_i^E from the main branch encoder and certainty pseudo-label Y_i^u are input into the uncertainty prototype module $UP(\cdot)$ to obtain the similarity map P_i^{pro} . Finally, it is mixed with the inverse transformed output $T^{-1}(P_i^t)$ from transform branch to generate a mixed pseudo label Y_i^{mix} . In each iteration, the network is initially trained with scribble annotations using scribble supervision loss ($\mathcal{L}_{ssl}$); consistency loss ($\mathcal{L}_{con}$) is used to minimize the difference between different outputs of the two branches; distance optimization loss ($\mathcal{L}_{dis}$) is used to optimize inter- and intra-class distances; pseudo-supervision loss ($\mathcal{L}_{seg}$) uses the generated mixed pseudo-label to further supervise network training; the boundary regression loss ($\mathcal{L}_{reg}$) is used to enhance boundary learning under the supervision of mixed pseudo-label. During the testing phase, we only use the output P^s of the main branch as the final prediction result.

Algorithm 1

The proposed method.

Input: X, Y ; // X : Original OCT B-scan; Y : Scribble Annotations corresponding to X .
Output: $B_m(\cdot), B_m^E(\cdot), B_t(\cdot)$ and $UP(\cdot)$ share the same parameter θ . // $B_m(\cdot)$: The main branch; $B_m^E(\cdot)$: The encoder of main branch; $B_t(\cdot)$: The transform branch; $UP(\cdot)$: The uncertainty prototype module.

- 1: **for** $i = 0, 1, \dots, n$ **do** // n iterations
- 2: get image and scribble annotation $\{X_i, Y_i\}_i \sim \{X, Y\}$; // get from a batch
- 3: $\{T(X_i), T(Y_i)\} \sim \{X_i, Y_i\}$; // get transformed image and scribble annotation with transform operation $T(\cdot)$
- 4: $P_i^s, P_i^r, Y_i^u \leftarrow B_m(X_i)$; // the segmentation probability map, the regression probability map and the certainty pseudo-label are generated by $B_m(\cdot)$
- 5: $P_i^t \leftarrow B_t(T(X_i))$; // the probability map is generated by $B_t(\cdot)$
- 6: $F_i^E \leftarrow B_m^E(X_i)$; // get the features from the encoder of the $B_m(\cdot)$
- 7: $P_i^{pro} \leftarrow UP(F_i^E, Y_i^u)$; // the similarity map is generated by $UP(\cdot)$
- 8: $T^{-1}(P_i^t) \sim P_i^t$; // obtained by the probability map from $B_t(\cdot)$ with the inverse transform operation $T^{-1}(\cdot)$;
- 9: $Y_i^{mix} \leftarrow 0.5 \times (P_i^{pro} + T^{-1}(P_i^t))$; // get the mixed pseudo label, see Eq. (13)
- 10: $\mathcal{L}_{ssl} \leftarrow -\nabla_{\theta} \frac{1}{b} \sum_{i=1}^b \mathcal{L}_{ssl}(P_i^s, P_i^r, Y_i, T(Y_i))$; // $\mathcal{L}_{ssl}$ denotes the scribble-supervised loss, see Eq. (21)

(continued on next page)

Algorithm 1 (continued)

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11:  $g_0^{con} \leftarrow -\nabla_{\theta} \frac{1}{b} \sum_{i=1}^b \mathcal{L}_{con}(P_i^s, P_i^{pro}, T^{-1}(P_i^t)); // \mathcal{L}_{con} \text{ denotes the consistency loss, see Eq. (22)}$ 
12:  $g_0^{dis} \leftarrow -\nabla_{\theta} \frac{1}{b} \sum_{i=1}^b \mathcal{L}_{dis}(P_i^{pro}, \text{argmax}(P_i^s)); // \mathcal{L}_{dis} \text{ denotes the distance optimization loss, see Eq. (12)}$ 
13:  $g_0^{seg} \leftarrow -\nabla_{\theta} \frac{1}{b} \sum_{i=1}^b \mathcal{L}_{seg}(P_i^s, Y_i^{mix}, T^{-1}(P_i^t)); // \mathcal{L}_{seg} \text{ denotes the pseudo-supervised loss, see Eq. (23)}$ 
14:  $g_0^{reg} \leftarrow -\nabla_{\theta} \frac{1}{b} \sum_{i=1}^b \mathcal{L}_{reg}(P_i^r, P_i^s, Y_i^{mix}); // \mathcal{L}_{reg} \text{ denotes the boundary regression loss, see Eq. (19)}$ 
15:  $\theta \leftarrow \theta + lSGD(\theta, g_0^{con} + \lambda(t)(g_0^{con} + g_0^{dis} + g_0^{seg} + g_0^{reg})); // \text{update the model parameters, see Eq. (24)}$ 
16: end for;
17:  $P^s \leftarrow B_m(X); // \text{the test stage}$ 

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Data availability

The authors do not have permission to share data.

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